

SPL: Orchestrating Workflows with Declarative Deterministic–Probabilistic Composition

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Abstract

Agentic workflows increasingly need to compose probabilistic LLM inference with deterministic, machine-checkable computation: plan a derivation, then verify it in a symbolic kernel; draft code, then test it against a specification. Declarative LLM-orchestration systems (LMQL, DSPy, and other recent designs) address the first need; symbolic tool-calling frameworks address the second; composing both today still requires hand-written glue code that is neither declarative nor portable. We present SPL (Structured Prompt Language), a declarative language that gives both computation modes - probabilistic (GENERATE/EVALUATE/WHILE) and deterministic (SOLVE/ASSERT) - first-class status in a single formal grammar sharing one variable namespace, so an LLM’s output flows directly into a symbolic kernel check and back without programmer-managed marshaling between two separately-orchestrated systems. A `.spl` specification only declares *what* workflow to run, not *which* model or verifier runs it - so the same script runs unchanged across various language models / backends / runtimes.

We validate SPL with an 80-recipe cookbook - including compilation to four external orchestration targets - and a controlled 1,320-run experiment on symbolic mathematics (SymPy backend; a same-scale Sage-backend experiment is reported separately in an appendix to avoid conflating backend familiarity with model capability). The solver arm’s kernel-execution pass rate spans 32–100% across eleven models, with the ranking among the top models indicative rather than statistically significant at this sample size; independently re-verifying each passing result by substituting it back into the original problem’s defining relation confirms 87.1% of these as semantically correct, isolating how much of the raw pass rate reflects execution success alone rather than a solved problem. SPL is released under Apache 2 license.

1 Introduction

1.1 The Problem: Fragmentation in LLM Programming

Building LLM-powered workflow systems today demands a daunting intersection of skills. A practitioner must master prompt engineering for effective LLM interaction, Python programming for orchestration logic, API wrangling across disparate model providers, and manual state management for multi-step reasoning. Each popular framework — LangGraph [30], AutoGen [1], CrewAI [2] — introduces its own abstractions, API surfaces, and execution models, creating a fragmented ecosystem where knowledge transfers poorly and vendor lock-in is the norm.

This situation echoes a pattern the data industry has seen before. In the early days of relational databases, programmers wrote imperative C and COBOL code to traverse data structures, manage cursors, and handle errors — all in application-specific ways. The introduction of SQL in the 1970s transformed this landscape

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by providing a declarative specification layer: users described *what* data they wanted, not *how* to retrieve it. This separation enabled decades of optimizer innovation underneath a stable surface language. The current state of LLM programming is analogous to the pre-SQL era: powerful capabilities buried under layers of imperative glue code.

The barrier to entry compounds the problem. To build even a simple self-refining workflow — one that drafts, critiques, and revises its own output — a developer must write 84–147 executable lines of Python across multiple framework abstractions.¹ This excludes the vast population of domain experts, analysts, and researchers who understand their workflows but lack the software engineering skills to express them in imperative orchestration code.

1.2 The Two-Mode Gap

All existing frameworks — declarative and imperative alike — operate in a single computation mode. LLM-centric systems (LangGraph, DSPy, AutoGen) produce approximate, non-reproducible outputs shaped by the model’s training distribution. Symbolic tools (SymPy, SageMath, Lean) produce exact, reproducible, machine-verifiable results. Neither world speaks the other’s language within a unified programming model. We call this the **two-mode gap**.

Real agentic tasks routinely require both. A homework assistant must plan a solution in natural language (probabilistic), execute each algebraic step exactly (deterministic), verify the result (deterministic), then explain it to the student (probabilistic). The coupling between modes may be *loose* — distinct phases passing data once — or *tight*: an **ASSERT** failure immediately triggers a corrective **GENERATE**, which revises the plan and re-enters the symbolic engine. Both patterns arise in practice; both require application-specific glue code to coordinate state transfer and error propagation at every mode crossing.

The gap is an expressiveness ceiling, not an engineering inconvenience. No existing workflow language lets the programmer declare “compute this step exactly” and “generate this step approximately” in the same specification. SPL closes it through *declarative composition*: the programmer specifies what each step computes and which mode it inhabits; the runtime resolves the boundary, manages state transfer, and enforces verification gates — without orchestration glue code. (Domain-support code — custom tool implementations backing a specific verifier or data source — is separate from orchestration and is not eliminated; §7.5 discusses this distinction.)

The relevant distinction is epistemic, not temporal — close to Kahneman’s dual-process theory [21], but sharper: Kahneman names the two systems *fast* and *slow*, conflating epistemic mode with cognitive speed, whereas a SymPy kernel may compute a derivative faster than the LLM does while a Lean 4 proof may take longer, with neither fact changing which mode each step inhabits. We name the modes directly instead: the *probabilistic* mode pattern-matches, generates hypotheses, and produces approximate outputs shaped by prior experience, regardless of wall-clock time; the *deterministic* mode applies explicit rules to produce exact, reproducible results, regardless of execution time. In SPL’s mapping, **GENERATE** and **EVALUATE** are probabilistic-mode primitives; **SOLVE** and **ASSERT** are deterministic-mode primitives. We use *probabilistic/deterministic* throughout the rest of this paper.

1.3 Our Contributions

We present SPL, a declarative language that composes both computation modes in a single coherent specification. The primary contributions of this paper are:

C1. Declarative two-mode composition. SPL provides **SOLVE** and **ASSERT** — primitives that route computation to a live Python kernel (IPython) rather than an LLM — alongside **GENERATE** and **EVALUATE**

¹Measured on the self-refine recipe (`cookbook/05_self_refine/`), which ships hand-built equivalents for four frameworks: AutoGen 84 executable LOC (118 lines total), CrewAI 90 (125 total), LangGraph 107 (158 total), PocketFlow 147 (225 total). Executable lines exclude blank lines and single-line comments; total lines include both. The equivalent `self_refine.spl` is 95 lines total, of which approximately 35 are executable SPL control-flow instructions; the remainder are three LLM prompt templates embedded as string literals (the `AS $$$...$$$` blocks) and blank lines. The 84–147 Python figures count inline prompt strings as executable lines, while the 35-line SPL figure excludes its prompt-template blocks; the comparison is therefore between orchestration-only SPL code and orchestration-plus-prompts Python code, not a like-for-like line count.

for the probabilistic mode. A single declarative specification freely composes both modes; variable bindings flow between them through a shared `@variable` namespace without programmer-managed marshaling. The mode boundary is visible in the source and managed entirely by the runtime. This declarative structure is not a virtue in itself: because the specification names only *what* each step computes, not *which* model or kernel executes it, the same `.spl` file can be retargeted across backends without modification — the capability C2 develops next.

C2. The DODA principle. Design Once, Deploy Anywhere: a single `.spl` specification is the invariant. The choice of model provider (Ollama, OpenRouter, Anthropic, Momagrid) and verifier (SymPy, SageMath, Lean) is resolved at invocation time via CLI flags, not embedded in the specification. The same file runs on a developer’s laptop and on a distributed inference grid without modification — declarative composition is the mechanism that makes this possible.

C3. Controlled empirical evaluation. A 1320-run experiment (11 models $\times$ 10 SymPy-backend problems $\times$ 2 arms $\times$ 6 repetitions; a same-scale Sage-backend experiment is reported in an appendix, kept separate from the main comparison to avoid conflating backend familiarity with model capability) on symbolic mathematics provides quantitative evidence of the two-mode composition benefit and surfaces a structured capability hierarchy: models that score 100% in the probabilistic arm (LLM-only math) may score as low as 32% in the two-mode arm (solver), not because they cannot solve the math but because they cannot produce the structured decomposition the deterministic engine requires. This reveals that format-compliance for declarative composition is a separable capability from mathematical reasoning.

C4. The verifier ladder. We implement a three-rung hierarchy of symbolic verifiers — SymPy (algebraic), SageMath (mathematical), Lean 4 (formal proof) — as ascending correctness guarantees composable into any SPL workflow via `ASSERT`. R1 and R2 are evaluated empirically in the 1320-run experiment (C3); R3 (Lean 4) is implemented with a working bridge (15/15 unit tests passing) and demonstrated as a design contribution, scoped outside this paper’s empirical evaluation. Each rung is a Python-callable tool registered at runtime; the workflow specification is agnostic to which rung is active. The ladder is itself declared, not coded: the escalation logic is `ASSERT` chains in `.spl`, not Python conditionals.

1.4 Paper Organization

Section 2 surveys related work, positioning SPL against imperative frameworks, declarative systems, and neurosymbolic AI. Section 3 presents the SPL language design — the full primitive set, semantics, and formal grammar. Section 4 formalizes the verifier ladder. Section 5 describes the implementation: how the two-mode executor dispatches probabilistic nodes to the adapter layer and deterministic nodes to the kernel layer, with model provider and verifier both selected at invocation time. Section 6 presents the 1320-run empirical evaluation (SymPy backend, $r=6$; the same-scale Sage-backend evaluation is in Appendix K). Section 7 discusses strengths, limitations, and future directions. Section 8 concludes.

2 Related Work

2.1 The Evolution of Data Programming

The history of data programming offers a clear precedent for the transition SPL proposes. In the 1970s, IBM’s System R project showed that a declarative query language (SQL) could match or exceed the performance of hand-coded navigational database access, because the declarative form exposed optimization opportunities invisible in imperative code [3]. By the 1980s, SQL had become the industry standard, and Oracle’s PL/SQL extension showed that procedural control flow (loops, variables, exception handling) could be layered on top of declarative queries without sacrificing the optimizer’s ability to reason about individual `SELECT` statements [4].

The parallel to LLM programming is direct. Today’s imperative frameworks — LangGraph’s state graphs, AutoGen’s conversation loops, CrewAI’s role-based agents — are the equivalent of pre-SQL navigational code: powerful, flexible, and impossibly fragmented. SPL proposes that the same declarative revolution can occur for LLM orchestration, following the same evolutionary arc: atomic declarative queries (`PROMPT`)

compose into procedural workflows (**WORKFLOW**) that an optimizer can eventually rewrite and route without altering the specification.

A critical difference from the SQL era: data queries are deterministic. LLM orchestration is not. This demands new primitives — **EVALUATE** for semantic branching, **WHILE** for quality-gated iteration, **EXCEPTION** for probabilistic failure modes — that have no SQL equivalent. And it demands a second computation mode entirely: **SOLVE** and **ASSERT** for the cases where a result must be exact rather than approximate.

2.2 Imperative Orchestration Frameworks

AutoGen [1] focuses on multi-agent conversation patterns where agents exchange messages. Its strength is natural dialogue flows; its limitation is that orchestration logic is embedded in Python and tightly coupled to the runtime.

CrewAI [2] introduces a role-based metaphor where specialized agents collaborate. It is intuitive for team-like workflows, but remains Python-first with no path to compilation or optimization.

PydanticAI takes a code-first approach, prioritizing developer ergonomics and type safety. Like the above, it operates exclusively in the probabilistic mode with no symbolic integration layer.

Semantic Kernel [33] pairs an LLM-driven planner with function-calling “skills,” overlapping conceptually with SPL’s **SOLVE/ASSERT**: a skill can wrap a deterministic computation. But the planner-to-skill boundary is a runtime tool-call contract, not a grammar construct — the mode split is not visible in a single declarative specification, and there is no formal verification gate analogous to **ASSERT**.

Haystack [34] composes typed pipeline components (retrievers, generators, rankers) for RAG and agentic search. Its typed component I/O is closer to SPL’s typed **INPUT/OUTPUT** declarations than most imperative frameworks, but the pipeline graph is Python-constructed, not an independent grammar, and it has no deterministic-kernel mode.

None of these frameworks provides a language-level mechanism to say “execute this step deterministically and verify the result before the workflow proceeds.”

2.3 Declarative LLM Systems

LMQL [5] pioneered SQL-like LLM interaction with output constraints (type, length, regex), achieving 26–85% cost reduction through constrained decoding. LMQL operates at the query level — individual prompts with output validation — and does not provide constructs for multi-step workflows or deterministic integration.

DSPy [6] introduces declarative modules (**Predict**, **ChainOfThought**) with a compiler that automatically optimizes prompts via few-shot bootstrapping. DSPy’s “declarative” is at the module-interface level: it optimizes which prompts to use, not how to orchestrate multi-step workflows. It provides no iteration, no exception handling, no symbolic integration, and no formal grammar. A worked contrast showing how the same self-refine task is expressed in DSPy assertions and LMQL token constraints versus SPL’s **EVALUATE/WHILE** runtime semantics is given in Appendix J.

SGLang [7] provides structured generation with runtime optimizations (RadixAttention, up to $6.4\times$ throughput). SGLang optimizes inference execution below the level of agentic patterns; it does not address workflow orchestration.

Agent Spec [8] defines a YAML/JSON schema for framework-agnostic agent specification. It is a configuration format, not an executable language, and lacks control flow, iteration, and exception handling within the specification itself.

eBay DSL [9] separates workflow specification from implementation across multiple backend languages, achieving 60% reduction in development time. It focuses on enterprise pipeline configuration and does not formalize LLM-specific primitives or symbolic integration.

Agentics 2.0 [10] introduces a logical transduction algebra where LLM calls are typed semantic transformations composable via algebraically grounded operators. It is the closest concurrent work in formalism, though algebraic rather than SQL-inspired, and it still operates exclusively in the probabilistic mode.

Compiled AI [11] addresses non-determinism through compile-time LLM invocation: the language model runs once at specification time to generate a deterministic executable artifact; no LLM is invoked at runtime. This achieves strong execution guarantees at the cost of losing all runtime adaptability — no **EVALUATE** branching, no **WHILE** quality gates, no **EXCEPTION** recovery. SPL takes the opposite stance: both modes remain live and equally accessible at runtime, interleaved through a single declarative specification. The tradeoff is explicit: SPL retains full probabilistic expressiveness (multi-shot LLM calls, adaptive routing, exception handling) alongside deterministic verification.

Blueprint First [12] proposes an engineering pattern that decouples workflow logic from the generative model. Expert-defined procedures are codified into an “Execution Blueprint” (deterministic engine), and the LLM is invoked only for bounded sub-tasks within it. While this echoes SPL’s DODA principle — the blueprint as invariant, the LLM as pluggable component — the approach remains a design pattern rather than an executable language. Blueprint First offers no symbolic verification layer (no **SOLVE/ASSERT** equivalent), no formal grammar, and no shared variable namespace allowing seamless data flow between modes. The blueprint is a manually maintained engineering artifact; SPL’s specification is the executable, compiled form.

PDL [27] is a YAML-based prompt format with Jinja2 variable templating and 15 block types (**model:**, **code:**, **for:**, **repeat:**, etc.). PDL’s design goal is structured, repeatable prompting: blocks render text in a document-oriented fashion, and **code:** blocks can invoke arbitrary Python. PDL does not define typed workflow variables, semantic branching conditioned on LLM output, formal exception handling, or a **SOLVE/ASSERT** verification layer; its evaluation consists of three qualitative case studies with no quantitative benchmark. SPL differs at the abstraction level: the unit of composition is a named, parameterized **WORKFLOW** with typed **INPUT/OUTPUT** declarations and first-class state persistence across **CALL** boundaries, not a document template. The two systems are complementary rather than competing: PDL excels at structured prompt authoring; SPL addresses multi-step workflow orchestration with formal correctness gates.

APPL [28] introduces a Python-embedded DSL for LLM programming that preserves Python control flow while providing prompt-construction helpers. APPL’s integration with Python is tighter than SPL’s (SPL is a separate language), giving more expressive power for ad-hoc scripting, but it provides no formal grammar, no **ASSERT**-based verification gate, and no symbolic solver integration.

VIEIRA [29] proposes a probabilistic-relational framework for programming with foundation models. Vieira treats language and vision models as stateless relational functions embedded in a Scallop datalog program; the logic substrate governs neuro-symbolic composition across language, vision, and database modalities. Unlike SPL, Vieira has no imperative workflow layer (no named procedures, **WHILE** repair loops, or **CALL** boundaries), no typed **INPUT/OUTPUT** declarations, and no hard **ASSERT** gate — its correctness mechanism is probabilistic relational inference rather than binary pass/fail verification.

Guidance [31] and **Outlines** [32] constrain LLM decoding to a target grammar or regex, guaranteeing structural output validity token-by-token. Both operate at the same level as SGLang’s structured generation — constraining a single generation call — with no multi-step workflow layer, no shared variable namespace across calls, and no deterministic-kernel integration; SPL’s **GENERATE** could adopt either as an implementation detail of a single probabilistic-mode step without changing the language.

2.4 LLM+Solver Composition

A distinct line of work explores coupling language models with symbolic solvers via mode composition: the LLM translates a natural language problem into a solver encoding, the solver executes deterministically, and the LLM translates the result back to natural language.

LLM+P [13] pioneered this pattern: language models generate PDDL planning encodings; a classical optimal planner (Fast Downward) solves them; the LLM narrates the solution. This foundational work established the “two-phase” pattern — probabilistic translation, deterministic execution, probabilistic in-

terpretation — that has influenced subsequent systems. The limitation is that the LLM is restricted to a translator role between fixed input/output schemas ($NL \leftrightarrow PDDL$); there is no multi-step workflow composition, shared variable namespace, or first-class language for orchestrating the two phases.

MCP-Solver [14] wraps constraint solvers (MiniZinc, PySAT, Z3, Clingo/ASP) as Model Context Protocol tools. The LLM builds a solver encoding through conversational tool calls, the solver executes, and the LLM interprets results. Like LLM+P, this uses a *tool-call pattern*: the LLM invokes the solver as a black box via a protocol-defined schema. There is no formal workflow language, no explicit mode boundary visible in source, and the variable-passing contract is implicit in tool schemas rather than explicit in a grammar.

DUPLEX [15] applies dual-system composition to robotic task planning: a lightweight “fast” system maps entity relations to PDDL via LLM; a classical planner executes; a “slow” system (heavier LLM with solver diagnostics) activates on failure. DUPLEX exhibits the $PLAN \rightarrow SOLVE \rightarrow ASSERT \rightarrow EXPLAIN$ lifecycle that SPL formalizes as a language pattern. However, DUPLEX is confined to the PDDL domain; modes do not share variables — they communicate through rigid schema — and the composition is hard-coded to the two-phase structure, not visible as a declarative grammar.

Across all three systems, the LLM is a component *within* a larger orchestration pattern, not a co-equal participant with explicit mode boundaries visible in source. SPL differs fundamentally: **SOLVE** and **ASSERT** are language primitives (not tool calls), the mode boundary is visible in the grammar, and variables flow between modes through a unified **@variable** namespace without schema marshaling. In a tool-call system, the developer writes explicit serialization/deserialization logic in Python to pass an LLM’s string output to a SymPy tool and parse the result back; in SPL, the **{@variable}** interpolation syntax makes this a language-level feature managed by the runtime.

2.5 Neurosymbolic AI

A growing body of work combines neural and symbolic computation, but at the model or algorithm level rather than the workflow language level.

AlphaGeometry [16] solves Olympiad geometry problems by alternating a neural language model (generating proof steps) with a symbolic deduction engine (verifying them). The neuro-symbolic loop is hard-coded to the geometry domain and implemented as a custom Python pipeline. SPL generalizes this pattern beyond geometry: the **SOLVE/ASSERT** primitives express the same loop declaratively for any domain with a Python-callable verifier, and the verifier (geometry engine, computer-algebra system, or any other symbolic tool) is a pluggable runtime choice rather than a hard-coded dependency.

Neural theorem proving systems (PACT, LeanDojo [17], HyperTree Proof Search [18]) use LLMs to generate proof tactics for a formal proof assistant, with the assistant itself as the verifier. SPL takes a different approach at the language level: rather than a fixed model-plus-proof-assistant pipeline, any symbolic verifier — a proof assistant included — is a pluggable **ASSERT** target selected at runtime, not hard-coded into the language.

Program synthesis systems (Codex [19], AlphaCode [20]) generate code and execute it to verify functional correctness. SPL’s **SOLVE** construct is the declarative equivalent of this loop: the programmer specifies the expression template; the kernel executes it; **ASSERT** verifies the result — without the programmer writing the execution and verification harness in Python.

Recent work on composable neuro-symbolic architectures offers insights into the two-mode approach. **Compositional AI Beyond LLMs** [23] examines from a systems/hardware perspective the characteristics of neuro-symbolic-probabilistic architectures, finding distinct memory and compute profiles and consistent performance gains over monolithic LLMs of comparable size. **Symbolic Seams** [24] proposes “symbolic seams” — explicit architectural breakpoints with typed boundary objects — as a design principle for composable systems, conceptually aligned with the mode boundary that SPL makes visible at the language level.

The common thread across these systems is that neuro-symbolic integration is implemented as bespoke application logic rather than as a language primitive. SPL’s contribution is to make that integration a first-class, reusable construct expressible in few keywords.

Table 1: System Comparison. Column definitions: *Grammar* — both computation modes (if the system has more than one) have first-class status in a formal grammar, not merely a documented convention; *Semantic Eval* — native LLM-as-judge branching on natural-language conditions (SPL’s `EVALUATE`), as opposed to string/regex matching only; *Both modes @runtime* — adaptive control flow (branching, iteration, exception handling) can operate across both modes during execution, not only at specification time; *Sym. Integration* — native, pluggable dispatch to a symbolic solver or verifier, as opposed to hand-written Python glue connecting the LLM layer to an external tool. Coding agents (e.g., Copilot, Cursor) are omitted from this table: they are powerful but opaque, proprietary tools with no declarative language exposed to program them, not specification languages comparable to the systems compared here.

System	Paradigm	Grammar	Semantic Eval	Both modes @runtime	Sym. Integration
LangGraph	Imperative	No	No	No	Manual (Python)
AutoGen	Imperative	No	No	No	Manual (Python)
CrewAI	Imperative	No	No	No	Manual (Python)
LMQL	Declarative	Yes	No	No	No
DSPy	Declarative	Partial (module sig.)	No	No	No
SGLang	Declarative	Yes	No	No	No
LLM+P	Two-phase	No	No	No	Partial (PDDL)
Blueprint First	Declarative	No	No	No	No
Compiled AI	Declarative	No	No	No	No
MCP-Solver	Tool-call	Partial (protocol schema)	No	Yes (ad hoc)	Via protocol
DUPLEX	Two-phase	No	No	No	Yes (PDDL domain)
Agent Spec	Declarative	Yes (schema)	No	No	No
eBay DSL	Declarative	Yes	No	Yes	No
Agentics 2.0	Algebraic	Yes (algebra)	No	No	No
AlphaGeometry	Neuro-sym.	No	No	Yes (hard-coded)	Yes (domain-specific)
SPL	Declarative	Yes (EBNF)	Yes	Yes (first-class)	Yes (pluggable)

2.6 Positioning Summary

SPL is the only system with **Yes** across all four differentiating columns in Table 1: (1) *Grammar* — both computation modes have first-class status in the formal EBNF (Appendix A), not just documented convention; (2) *Semantic Eval* — `EVALUATE` provides native LLM-as-judge branching on natural-language conditions; (3) *Both modes @runtime* — adaptive control flow (`EVALUATE`, `WHILE`, `EXCEPTION`) operates across both modes during execution, not only at specification time; (4) *Sym. Integration* — `SOLVE/ASSERT` provide native, pluggable dispatch to a symbolic solver, not hand-written Python glue. A fifth property that Table 1 does not tabulate as a column, because no comparison system offers a partial case worth contrasting, is SPL’s unified `@variable` namespace: outputs of one mode are available as inputs to the other without tool-schema marshaling. The critical distinction from every other declarative system is that SPL’s declarativity extends across the mode boundary: the `.spl` source specifies composition of both modes; no Python glue bridges them.

3 The SPL Language

3.1 Design Principles

SPL is governed by the following design principles:

P1. Declarative composition over imperative implementation. The programmer specifies *what* each step computes and *which mode* it inhabits — not which model, which verifier, or which execution path achieves it. The two modes may be woven loosely (distinct probabilistic and deterministic phases passing data between them) or tightly (fine-grained interleaving where `ASSERT` failure immediately re-enters `GENERATE`); SPL handles both ends of this spectrum with the same primitives. The runtime resolves the mode boundary; the `.spl` source is the invariant across both coupling styles, both model providers, and both deployment environments.

P2. Two-mode primitives as co-equal first-class constructs. SPL provides primitives for both computation modes — probabilistic (intuitive, approximate) and deterministic (mechanical, exact), distinguished epistemically rather than temporally (see §1.2). The probabilistic mode (`GENERATE`, `EVALUATE`, `WHILE`, `EXCEPTION`) addresses the characteristics of LLM computation: approximate, non-reproducible, quality-assessed, cost-sensitive. The deterministic mode (`SOLVE`, `ASSERT`) addresses the characteristics of symbolic computation: exact, reproducible, machine-verifiable. Neither mode is subordinate to the other in the language design — their composition in a single specification is the defining property of SPL.

P3. Shared variable space. Outputs of `GENERATE` steps are available as inputs to `SOLVE` steps, and vice versa, through a single `@variable` namespace. The programmer does not marshal values between modes; the runtime manages the boundary. Two syntactic forms govern variable use: `@name TYPE` is the *declaration/binding* form (used in `INPUT/OUTPUT` signatures, `SOLVE` targets, and assignment), while `{@name}` is the *interpolation/evaluation* form (used inside `PROMPT` templates, `SOLVE` expressions, and `ASSERT` predicates). The executor substitutes each `{@name}` reference with the current value of `@name` before dispatch. This is the mode-crossing seam, and it is the same mechanism as SPL f-strings (`f"..."`) applied uniformly across all string contexts.

P4. DODA (C2, §1.3). At the language-design level, DODA is realized by keeping `--adapter`, `--model`, and `--kernel` invocation-time flags, never spec-embedded choices.

P5. Compilation target. The language is designed to produce an optimizable intermediate representation (IR), enabling future optimizer passes (`GENERATE` merging, `SELECT` caching, model routing) without changing the surface language.

3.2 The SQL Analogy

SPL’s declarative form is not an independently-argued virtue borrowed from SQL; it is the mechanism that makes DODA (C2, §1.3) possible. A specification that declares only *what* each step computes, not *which* model or kernel executes it, can be retargeted across backends without modification — demonstrated directly by the same unmodified `.spl` recipe running to closely-agreeing outcomes on two independently-implemented runtimes (§5). The SQL and PL/SQL stack below is offered only as a mental-model aid for reading SPL’s syntax, not as the argument for why declarative form matters here:

- **SELECT** assembles the context that flows *into* the LLM — analogous to `SELECT ... FROM ... WHERE` gathering rows from a table.
- **GENERATE** invokes the LLM and captures what flows *out* — analogous to the result set returned by a SQL query.
- **PROMPT** binds `SELECT` and `GENERATE` into a named, independently executable unit — analogous to a single SQL query.
- **WORKFLOW** adds variables, loops, branching, and exception handling around `PROMPT` calls — analogous to PL/SQL layering procedural control flow over SQL queries.
- **SOLVE** routes a Python expression to the deterministic kernel — an extension that has no SQL analogue.

This last point marks the boundary of the SQL analogy. SPL must go further because LLM orchestration requires a second computation mode that relational data access does not. A second boundary deserves acknowledgment: SQL optimization rests on relational algebra equivalences over deterministic set operations, whereas **GENERATE** samples from a probability distribution. Future SPL optimizations (batching, caching, model routing) must therefore operate at the *workflow* level — reordering independent nodes, fusing adjacent calls — rather than applying algebraic rewrites to individual **GENERATE** invocations.

3.3 Deterministic Primitives

3.3.1 SOLVE — Kernel-Routed Computation

SOLVE dispatches a computation to the selected verifier engine and assigns the result to a workflow variable. Its behavior is backend-dependent: for the SymPy and SageMath rungs (R1/R2), it evaluates a Python expression in the live IPython kernel, as shown below; for the Lean 4 rung (R3), it instead drives a multi-stage formalize/typecheck/judge-faithfulness/prove protocol rather than evaluating a single expression. The surface syntax is unchanged across rungs, but the underlying computation model differs by rung; see §4.4 for the Lean case.

```
SOLVE @derivative SYMPY := "diff({@expression}, x)"
```

The {@variable} syntax binds any @variable currently in scope into the expression before dispatch. This is the boundary-crossing mechanism: the LLM’s output (held in @expression) becomes the input to the symbolic engine without programmer-managed marshaling. The result (@derivative) is available immediately to subsequent **GENERATE** or **SOLVE** steps. The substitution is performed by the executor via regex matching on the parsed token stream — the runtime resolves each {@identifier} reference against the variable store before constructing the kernel expression. The **SOLVE** kernel (SymPy/SageMath/Lean) runs in an out-of-process IPython subprocess; **CREATE TOOL_API** code executes via `exec()` within an isolated in-process namespace containing only the declared parameters and Python stdlib. Malformed expressions raise a `SyntaxError` or `NameError` caught by the `EXCEPTION` handler.

SOLVE is mode-explicit by design. The programmer cannot accidentally invoke the symbolic engine from a **GENERATE** call or the LLM from a **SOLVE** call. The mode boundary is visible in the source.

3.3.2 ASSERT — Verification Gate

ASSERT evaluates a Boolean predicate and gates workflow continuation on the result. The executor dispatches to one of two paths depending on whether a kernel session is active: (1) the IPython kernel, for arbitrary Python expressions that require imports or a live computation environment (e.g., symbolic math, graph libraries); (2) the registered **TOOL_API** namespace, for predicates that call a registered tool directly — these execute without a kernel and require no `-kernel` flag:

```
-- Kernel path: symbolic math expression (requires --kernel)
ASSERT simplify({@sympy_result} - diff({@expression}, x)) == 0
    OTHERWISE RETURN @explanation WITH status = 'verification_failed'

-- Tool-API path: registered TOOL_API predicate (no kernel required)
ASSERT is_optimal(@solution)
    OTHERWISE RETURN @report WITH status = 'infeasible'
```

On success, the workflow continues — the claim is machine-verified. On failure, the **OTHERWISE** body executes, typically committing a result with an error status or escalating to a higher verifier rung. **ASSERT** is the mechanism by which SPL workflows achieve correctness guarantees: a step that passes **ASSERT** is proven correct by the deterministic predicate, not merely judged plausible.

It is important to distinguish two failure modes that **ASSERT** separates: (a) *execution failure* — the generated symbolic plan raises a `SyntaxError` or `NameError` in the kernel, indicating the LLM produced structurally

invalid code; and (b) *semantic failure* — the plan executes without error but **ASSERT** returns false, indicating the plan encodes an incorrect transformation (e.g., the derivative of a wrong expression). The solver-arm pass criterion in §6 counts a cell as passing only when the kernel executes all steps *and* the final **ASSERT** `verify({@result}, {@problem})` returns true. Execution success alone is not a pass. This distinction is the core of what “machine-verified correctness” means in this paper: the **ASSERT** predicate is the ground-truth oracle, not a proxy for successful execution. Because **ASSERT** accepts any Python Boolean expression, tolerance-based assertions for numerical computation are naturally supported (e.g., **ASSERT** `abs({@result} - {@expected}) < 1e-6`); the language imposes no restriction to exact symbolic equality.

The deterministic primitives are not limited to mathematics. Any Python-evaluable predicate works as an **ASSERT** condition:

```
-- Schema validation: verify LLM output conforms to expected JSON structure
ASSERT json.loads({@api_response}) and validate(json.loads({@api_response}), schema)
    OTHERWISE RETURN @result WITH status = 'schema_violation'

-- Length constraint: ensure generated content meets minimum requirements
ASSERT len({@report_body}) >= 500
    OTHERWISE RETRY

-- Graph property: verify generated network satisfies connectivity invariant
ASSERT nx.is_connected(G) and nx.number_of_nodes(G) == {@expected_nodes}
    OTHERWISE RETURN @graph WITH status = 'graph_invalid'
```

These examples illustrate that **SOLVE/ASSERT** form a general *deterministic computation mode*, not a math-specific feature. **SOLVE** always routes to the IPython kernel. **ASSERT** dispatches to the kernel for arbitrary Python expressions, or directly to the registered tool namespace when the predicate is a **TOOL_API** call — enabling formal gates without a kernel session. Any domain with a Python-callable verifier — JSON schema validation, unit checking (pint), constraint solving (Z3), statistical testing (scipy), LP/MIP solvers (PuLP) — plugs into the same primitives. We label this mode “deterministic” throughout the paper to emphasize its reproducibility and machine-verifiability relative to LLM inference. Strictly, the Python kernel can execute non-deterministic code (e.g., MCMC sampling, randomized algorithms); in such cases the **ASSERT** gate still enforces a verifiable postcondition on the result, preserving the mode’s role as the verification boundary regardless of whether the kernel computation itself is stochastic.

3.3.3 CALL PARALLEL — Concurrent Branch Dispatch

```
CALL PARALLEL
    CALL arm_solver(@problem) INTO @result_solver
    CALL arm_llm_only(@problem) INTO @result_llm
END
```

Branches execute concurrently via `asyncio.gather` on a single node, or route to distinct Momagrid worker nodes on a distributed grid. This construct makes A/B experiments — deterministic arm vs. probabilistic arm — a language-level primitive rather than a test-harness concern. When parallel branches contain **SOLVE** or **ASSERT** steps, each branch operates on an isolated copy of the kernel namespace to prevent race conditions; results are merged into the parent scope only upon branch completion via the **INTO** binding.

3.4 Probabilistic Primitives

3.4.1 SELECT and GENERATE — The Atomic I/O Contract

SELECT assembles the prompt from multiple context sources; **GENERATE** invokes the LLM and captures the result:

```
WORKFLOW explain_concept
```

```

INPUT: @concept TEXT, @audience TEXT DEFAULT 'undergraduate'
OUTPUT: @explanation TEXT
DO
  GENERATE explanation(concept, audience) INTO @explanation
  SELECT
    system_role('You are a clear, precise instructor.'),
    @concept AS concept,
    @audience AS audience
  PROMPT "Explain {@concept} to a {@audience} student in 3 sentences."
END

```

In the common case shown above, each `AS` clause aliases a variable to itself (`@concept AS concept`) — `SELECT`'s load-bearing job here is assembling heterogeneous context sources (`system_role(...)` alongside workflow variables) into a single named binding set for `PROMPT` interpolation, not renaming. `AS` earns its keep when a source and its target name genuinely differ (e.g., selecting from a computed sub-expression or a value bound under a different name inside a `CALL` frame), but that case is not exercised in this listing and we do not claim otherwise. `SELECT/AS` was introduced in SPL v1.0, before `CREATE FUNCTION` (Appendix A) existed as a distinct construct for named, parameterized, reusable prompt templates; for the common case of explicit prompt engineering, `CREATE FUNCTION` is now the more direct mechanism, and `SELECT/AS` remains in the language for inline, one-off context assembly rather than as the primary prompt-engineering surface.

3.4.2 EVALUATE — Semantic Branching

`EVALUATE` uses the LLM as a runtime judge for conditions that cannot be expressed as deterministic comparisons:

```

EVALUATE @explanation
  WHEN contains('undefined') OR contains('undefined symbol') THEN
    @explanation := @explanation + '\n\n[Note: terms defined in appendix]'
  WHEN 'the explanation is unclear or contains unexplained jargon' THEN
    @status := 'needs_revision'
  WHEN = 'satisfactory' THEN
    @status := 'complete'
  ELSE
    @status := 'needs_revision'
END

```

Three condition forms coexist here, each dispatched differently: `contains(...)` is a deterministic string-containment check; the free-form quoted string `'the explanation is unclear or contains unexplained jargon'` is a natural-language predicate with no deterministic reduction, dispatched to the LLM judge — this is the branch that demonstrates genuine semantic evaluation, since no exact-match or substring test could implement it; and `= 'satisfactory'` is a literal equality comparison, evaluated deterministically against a prior exact-match value (e.g., a status field set by an earlier step), not against `@explanation`'s free text. This hybrid detection rule means the programmer does not annotate which branches are semantic and which are deterministic — the parser infers it from syntax: quoted natural-language conditions go to the judge, containment/equality checks are evaluated directly.

3.4.3 WHILE — Quality-Gated Iteration

```

WHILE 'explanation is incomplete or unclear' DO
  GENERATE refine_explanation(@explanation, @feedback) INTO @explanation
END

```

The termination condition is evaluated by the LLM on each iteration: the executor assembles all current `@variable` values as context and asks the LLM judge whether the condition still holds. A built-in maximum iteration guard (configurable, default 15) prevents infinite loops.

3.4.4 EXCEPTION — LLM-Aware Error Handling

```

EXCEPTION
  WHEN HallucinationDetected THEN
    RETURN @result WITH status = 'hallucination_detected'
  WHEN ContextLengthExceeded THEN
    @problem := summarize(@problem)
    RETRY
  WHEN BudgetExceeded THEN
    RETURN @partial_result WITH status = 'budget_exceeded'
END

```

The exception taxonomy formalizes failure modes specific to LLM computation — hallucination, refusal, context overflow, budget violation — that have no equivalent in conventional exception hierarchies. Detection mechanisms vary by type: `ContextLengthExceeded` and `BudgetExceeded` are raised deterministically by the adapter when API limits are hit; `ModelRefused` is detected by pattern matching on refusal templates in the response. `HallucinationDetected` is currently a proposed exception type — the runtime does not yet implement automatic hallucination detection; the programmer can raise it explicitly via an `EVALUATE` guard that dispatches to an LLM judge. The taxonomy is designed to be extensible as detection capabilities mature.

3.5 The Two-Mode Composition Pattern

Real agentic tasks follow a recurring four-stage lifecycle: **PLAN** (probabilistic mode decomposes the problem), **SOLVE** (deterministic mode executes it exactly), **ASSERT** (deterministic mode verifies the result), **EXPLAIN** (probabilistic mode narrates it in natural language). The following example shows this lifecycle expressed declaratively in the following SPL workflow:

```

WORKFLOW solve_with_verification
  INPUT: @problem TEXT
  OUTPUT: @explanation TEXT
DO
  -- [PLAN -- Probabilistic] LLM decomposes the problem
  GENERATE decomposition(@problem) INTO @steps_text
  SELECT @problem AS problem
  PROMPT 'Decompose this problem into symbolic computation steps,
         one per line in the format: expression|operation'

  -- [PLAN -- Probabilistic] Sanity gate before kernel handoff
  GENERATE plan_check(@steps_text, @problem) INTO @plan_verdict
  SELECT @steps_text AS plan, @problem AS problem
  PROMPT 'Does this plan correctly address the problem? Reply: pass or fail'

  EVALUATE @plan_verdict
  WHEN = 'fail' THEN
    RETURN @explanation WITH status = 'plan_sanity_error'
  END

  -- [SOLVE -- Deterministic] Kernel executes each symbolic step exactly
  CALL solve_chain(@steps_text) INTO @verified_result

  -- [ASSERT -- Deterministic] Verify the chain answer against the problem
  ASSERT verify({@verified_result}, {@problem})
  OTHERWISE RETURN @explanation WITH status = 'verification_failed'

  -- [EXPLAIN -- Probabilistic] LLM narrates the verified result

```

```

GENERATE explanation(@verified_result, @problem) INTO @explanation
  SELECT @verified_result AS result, @problem AS problem
  PROMPT "Explain this verified solution in clear prose: {@result}"

RETURN @explanation WITH status = 'complete'
END

```

This 30-line workflow composes five mode transitions declaratively: probabilistic decomposition $\rightarrow$ probabilistic sanity gate $\rightarrow$ deterministic execution $\rightarrow$ deterministic verification $\rightarrow$ probabilistic narration. Each transition is expressed in SPL syntax; no Python bridges the modes. (Both `solve_chain()` on the `CALL` line and `verify()` on the `ASSERT` line are user-registered Python callables — the former dispatching the decomposed steps to the symbolic kernel, the latter checking the result — not built-in SPL primitives; see Appendix I for implementation patterns and a source-code reference for both.)

The `{@var}` template syntax is the *formalization boundary* of SPL: the seam at which probabilistic-mode outputs (unstructured text from `GENERATE`) are cast into typed expressions consumed by the deterministic mode (`SOLVE/ASSERT`). The programmer does not write serialization code; the runtime resolves variable bindings before kernel dispatch. The mode crossing is visible at the source level — every `{@...}` reference marks a probabilistic-to-deterministic handoff — and is managed entirely by the language.

The programmer writes the same `.spl` file regardless of which LLM and which symbolic engine execute at runtime — the declarative specification is the invariant, and the DODA principle holds across both the model axis and the verifier axis.

3.6 Formal Grammar

The SPL grammar defines the following key production rules for two-mode integration:

```

solve_stmt      ::= 'SOLVE' '@'identifier ['SYMPY' | 'SAGE' | 'LEAN'] ':= ' string_expr
assert_stmt     ::= 'ASSERT' python_expr ['OTHERWISE' stmt_block]
parallel_stmt   ::= 'CALL' 'PARALLEL' (call_stmt)+ 'END'
stmt            ::= ... | solve_stmt | assert_stmt | parallel_stmt

```

The `SYMPY | SAGE | LEAN` type annotation on `SOLVE` is optional; when omitted, the runtime selects the kernel registered at invocation time. The full grammar is provided in Appendix A.

4 The Verifier Ladder

4.1 Motivation: Ascending Certification Scope

LLM-generated mathematical reasoning is probabilistic by nature: a model scoring 95% on a benchmark is wrong on 1 in 20 problems, with no signal to the caller about which ones. Symbolic verification closes this gap — but symbolic verification itself is not monolithic. A SymPy check of polynomial differentiation takes milliseconds and requires no external tooling. A formal Lean 4 proof of the same claim takes seconds, requires Mathlib, and produces a machine-checkable certificate that any Lean installation can re-verify independently. The two checks are not interchangeable: one is appropriate for routine computation, the other for publication-quality assertions. **The rungs differ in what class of claim each can certify, not in how trustworthy the engine is within that class:** a SymPy identity check and a Lean proof are each exactly correct for the claims they are built to check — escalating a rung is a decision to seek a stronger form of certificate, not a correction of a less reliable one.

The SPL verifier ladder makes this trade-off explicit and programmable. Three rungs address different cost/confidence operating points, ordered by the scope of what they certify rather than by reliability:

Rung Engine			What it certifies	Setup	Typical use
R1	SymPy (python3 kernel)		Algebraic identity, calculus, matrix operations	<code>pip install sympy</code>	Routine STEM computation
R2	SageMath (sagemath kernel)		Number theory, geometry, polynomial rings, combinatorics	<code>pip install 'spl-llm[sage]'</code>	Deeper mathematical claims
R3	Lean (lean_bridge)	4	Machine-checked formal proof against Mathlib	<code>bash setup_lean.sh --with-mathlib</code>	Publication-quality assertions

A workflow selects a rung via the `--kernel` and `--param backend=` flags at invocation time, optionally probing multiple rungs in parallel via `CALL PARALLEL`, with each rung receiving an engine-appropriate expression for the same underlying claim, to find the highest rung that certifies it. Expression syntax is rung-specific — a SymPy call, a Sage call, and a Lean statement are written differently for the same claim, as §4.5 shows — but the *workflow structure* (which rung to try, in what order, with what fallback) is identical across all three rungs. The DODA principle applies to verifier selection just as it applies to model selection: the orchestration logic, not the expression text, is what stays invariant.

4.2 Rung R1: SymPy

The SymPy rung runs inside an IPython kernel started with the standard `python3` kernelspec. Seventeen mathematical operations are registered as SPL tools via `CREATE TOOL_API`:

```
CREATE TOOL_API solve_step_with_sympy(expression, operation)
'''
import sympy as sp
from sympy.abc import x, y, z, t, n
ops = {
    "diff":      lambda e: sp.diff(e, x),
    "integrate": lambda e: sp.integrate(e, x),
    "solve":     lambda e: sp.solve(e, x),
    "eigenvalues": lambda e: sp.Matrix(eval(e)).eigenvals(),
    "dsolve":    lambda e: sp.dsolve(sp.sympify(e)),
    ...    -- 12 further operations
}
result = ops[operation](sp.sympify(expression))
return f"{result}|{sp.latex(result)}"
'''
```

The tool returns a `bare_result|human_readable` protocol: the bare result is available as a string for subsequent `{@var}` interpolation; the human-readable side is written to the chain trace log. On failure the sentinel `solver_error|...` is returned, which the workflow detects via `EVALUATE` and exits with an explicit status rather than propagating a bad value downstream.

4.3 Rung R2: SageMath

SageMath extends SymPy’s algebraic reach into number theory (Galois groups, elliptic curves, modular arithmetic), differential geometry, and combinatorics. The runtime discovers the SageMath Jupyter kernelspec automatically and raises a clear install error if absent (see Appendix C for setup). The same `SOLVE / ASSERT` primitives and `{@var}` interpolation work unchanged — the only runtime difference is which kernelspec launches. This is the DODA principle applied at the verifier level: the workflow programmer writes identical SPL regardless of which mathematical engine executes beneath.

4.4 Rung R3: Lean 4

The Lean rung targets a qualitatively different class of claim: a machine-checked formal proof against Mathlib rather than an algebraic identity, via a four-stage formalize/typecheck/judge-faithfulness/prove protocol (full SPL source in Appendix G.5) that returns a *badge*: `machine_proved`, `statement_checked`, `unfaithful`, or `unverified`. The Lean kernel, not the LLM, certifies the Typecheck and Prove stages; a failed proof never blocks delivery, only withholds the higher badge, so the rung can run speculatively in a `CALL PARALLEL` branch alongside a SymPy arm that already delivers an answer.

One stage remains probabilistic: `judge_faithfulness`, a second LLM call comparing the formalized Lean statement against the original claim, is the sole non-kernel-certified gate in the protocol — typechecking and provability alone cannot catch a well-formed, provable Lean statement that still misrepresents what it was asked to formalize. `machine_proved` therefore certifies the proof *for the statement the judge accepted as faithful*, not that the judge was correct to accept it; this paper reports no independent accuracy estimate for that judge (extended discussion, including the design rationale and cost/setup trade-offs across rungs, in Appendix G.5). As stated in §1.3 (C4) and §6.1, R3 is a working design contribution (15/15 unit tests passing) and is not part of the 1,320-run empirical evaluation. One scoping note on DODA: the formalization stage currently uses a general-purpose model through an ordinary `GENERATE` call, so the “same spec, any model” claim holds as implemented; were a fine-tuned NL→Lean translator introduced for this stage, it would be a fixed-translator configuration for that step specifically — outside the general DODA claim rather than a violation of it, the same way a hard-coded prompt template would be.

4.5 ASSERT as the Inter-Rung Gate

ASSERT is the mechanism by which the verifier ladder enforces correctness within a workflow. Each rung produces a value that can be checked, and ASSERT failure triggers escalation:

```
-- Try R1 first (inside a workflow body)
SOLVE @derivative SYMPY := "diff({@expression}, x)"
ASSERT simplify({@derivative} - diff({@expression}, x)) == 0
  OTHERWISE
    -- R1 could not verify; escalate to R2
    SOLVE @derivative SAGE := "SR({@expression}).diff(x)"
    ASSERT bool({@derivative} == diff({@expression}, x))
      OTHERWISE RETURN @explanation WITH status = 'r2_failed'
```

This escalation pattern is idiomatic SPL declarative composition: try the cheapest rung, gate on ASSERT, compose with a higher rung on failure. No Python glue code is required; the entire escalation logic — including the mode transitions between LLM narration and kernel verification — is expressed declaratively in the `.spl` source. The hand-written equivalent requires the programmer to manage kernel sessions, exceptions, and result marshaling explicitly:

```
sympy_kernel = start_kernel("python3")
try:
    derivative = sympy_kernel.execute(f"diff({@expression}, x)")
    if not sympy_kernel.execute(
        f"simplify({@derivative} - diff({@expression}, x)) == 0"
    ):
        raise VerificationError("r1_failed")
except (KernelError, VerificationError):
    sage_kernel = start_kernel("sagemath")
    try:
        derivative = sage_kernel.execute(f"SR({@expression}).diff(x)")
        if not sage_kernel.execute(
            f"bool({@derivative} == diff({@expression}, x))"
```

```

        ):
            return {"status": "r2_failed"}
    except KernelError:
        return {"status": "r2_failed"}
    finally:
        sage_kernel.close()
finally:
    sympy_kernel.close()

```

Two kernel lifecycles, two exception paths, and manual result marshaling between Python and each kernel’s native representation replace the six-line `.spl` block above — the SPL runtime manages the kernel-session and exception-handling code implicitly, so line count is only part of what changes.

4.6 Pluggability

The verifier ladder is open-ended: any Python-callable verifier plugs in without modifying the language or runtime. A workflow author registers a new tool via `CREATE TOOL_API`, selects it via a parameter or `--kernel` flag, and writes `ASSERT` conditions against its output. Beyond the three rungs demonstrated in this paper, the pattern generalizes to: unit verification (`pint`), graph property checking (`NetworkX`), constraint solving (`Z3`), type checking (`mypy`), or statistical hypothesis testing (`scipy.stats`). The language makes no assumption about what the kernel computes — only that it is Python-callable and that the result is a string the workflow can inspect.

5 Implementation

The executor parses a `.spl` source into an AST and dispatches each node to one of two engines: probabilistic nodes (`GENERATE`, `EVALUATE`, `WHILE`, `EXCEPTION`) route to the adapter layer, which normalizes calls across 14 LLM providers behind a two-method interface (`generate` / `generate_multimodal`). Deterministic nodes (`SOLVE`, `ASSERT`) route to the deterministic layer. `SOLVE` always dispatches to a persistent out-of-process Jupyter kernel for symbolic computation. `ASSERT` dispatches to the kernel when a kernel session is active; otherwise it evaluates the predicate directly in the registered `TOOL_API` namespace, enabling formal gates without a kernel session. The kernel activates only when `SOLVE` is encountered or `--kernel` is specified; otherwise no kernel process is started. Adapter and kernel are both selected at invocation time (`--adapter`, `--model`, `--kernel`) and never named in the `.spl` source — the deterministic mode of DODA mirrors the probabilistic one.

The execution pipeline routes `GENERATE/EVALUATE` to the adapter layer, `SOLVE` to the kernel, and `ASSERT` to either the kernel or the registered tool namespace depending on whether a kernel session is active, with all modes sharing state for seamless mode crossing without programmer-managed marshaling code.

The two modes share state through `{@var}` interpolation: before a `SOLVE` expression is dispatched to the kernel, the executor inlines current SPL variable values into the expression string. The result is captured and bound back to a `@var`, making it available to subsequent `GENERATE` or `EVALUATE` nodes. This variable-passing contract is the integration seam between the two modes; no explicit marshalling code is required in the workflow.

The declarative form does more than offer syntactic convenience: it is designed to *enable* optimizations that are impossible in imperative orchestration code. A static optimizer is already implemented and shipped (`spl.optimizer.Optimizer`, exposed via `spl3 explain <file>.spl`): it walks the AST without making any LLM call and produces a per-PROMPT/WORKFLOW execution plan — token-budget allocation across `SELECT` sources, priority ordering, proportional compression under budget pressure, and cost estimation — which a hand-written Python orchestration script has no equivalent surface to analyze, because its LLM calls and control flow are interleaved in ordinary code rather than declared as data.

This static-planning capability is mirrored in the independent Go runtime (<https://github.com/digital-duck/SPL.go>, [internal/executor/planner.go](https://github.com/digital-duck/SPL.go/blob/main/internal/executor/planner.go)), giving two independent implementations agree-

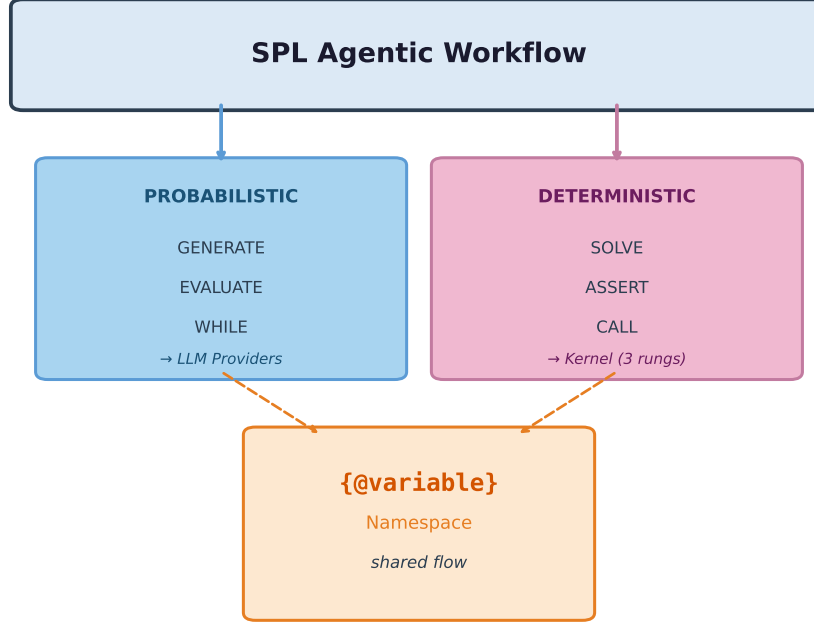


Figure 1: Execution Pipeline: The executor dispatches probabilistic nodes to the adapter layer (14 LLM providers) and deterministic nodes to a pluggable kernel, with both modes sharing state through the `{@variable}` namespace.

ment on what the IR means. The two runtimes are validated not only at this static-planning level but at full experimental scale: the identical `.spl` neurosymbolic recipe used for the results in §6 was run to completion on both — ten models (the original `r=3` roster, predating the `qwen3` addition described in §6.2), twenty problems, two solver modes, three repetitions, 1,200 cells each, no source changes between runtimes — yielding closely agreeing pass rates (87.2% Python vs. 87.9% Go) and round-trip-verified rates (88.3% vs. 88.8% of Pattern-1-pass runs). `qwen3` was subsequently run on both runtimes at the same scale (20 problems, two arms, three repetitions, 120 cells each) — with `qwen3`’s default thinking trace active — and shows the same agreement at the solver-arm pass-rate level: 96.7% on both Python and Go (58/60 cells each, identical count). This pass-rate parity result is not expected to change under the thinking-disabled configuration used for the main results (§6.2): both runtimes dispatch the identical plan to the identical kernel, and thinking mode had no material effect on `qwen3`’s Python-runtime pass rate either (§6.2), so a mechanism by which thinking mode alone would move Go-runtime pass rate specifically is not apparent. Round-trip verification for `qwen3` diverges more between runtimes than the original ten-model comparison, however: of the 58 Pattern-1-pass cells, Python logs all 58 as `ROUNDTRIP_PASS`, while Go logs 30 `ROUNDTRIP_PASS`, 2 `ROUNDTRIP_FAIL`, and 26 `UNPARSEABLE` — a 44.8% unparseable rate, well above the 4.6% baseline reported for the main pooled comparison (§6.2). Restricting to parseable cells, Go’s round-trip-verified rate is 93.8% (30/32) vs. Python’s 100% (58/58); we flag this as an open discrepancy rather than a settled finding. Unlike the pass-rate parity above, this specific gap plausibly *is* thinking-mode-dependent — it may reflect a Go-runtime round-trip-logging gap specific to `qwen3`’s extended-thinking output format (in which case re-running this comparison with thinking disabled would be expected to close it) or a smaller-sample artifact; either way it warrants dedicated investigation before being treated as a runtime-parity regression.

What remains future work (§7.7) is *runtime* rather than static optimization: fusing adjacent `GENERATE` calls into a single batched API request, caching `SELECT` results across workflow runs, reordering independent `SOLVE` steps for kernel throughput, and routing cheap sub-tasks to smaller models — all without modifying the `.spl` source. An imperative Python script that interleaves LLM calls with control flow cannot be rewritten by either kind of optimizer without understanding the programmer’s intent; a declarative specification makes

that intent explicit in the IR, which is what both the shipped static optimizer and the future runtime optimizer operate on. The separation of specification (`.spl`) from execution (adapter/kernel) is what makes this possible: the runtime is free to choose *how* to execute because the source only declares *what* to execute.

The same `.spl` source compiles to idiomatic code in four target frameworks via `sp13 splc (--target langgraph, go, typescript, pocketflow)`. The compiler reads the AST directly and maps each construct to the target’s natural idiom — `StateGraph` nodes for LangGraph, goroutines for Go parallel branches, `Promise.all` for TypeScript concurrency, and the `prep/exec/post ETL node` pattern for PocketFlow. This cross-framework compilation demonstrates that the declarative AST is a genuine intermediate representation, not a Python-specific wrapper: the same workflow specification produces native asynchronous code in each target’s natural concurrency model. Full compilation examples and runtime details (kernel substrates, adapter bootstrap, template resolution, ecosystem tooling) are provided in Appendices B, F, and G.

PocketFlow recipe migration. PocketFlow [26] is a minimalist open-source Python framework for LLM orchestration that organizes computation as directed graphs of ETL nodes (`prep/exec/post`). SPL supports the reverse direction as well as forward compilation: existing PocketFlow Python recipes can be migrated to `.spl` via a two-phase pipeline — (1) an LLM reads the Python source and writes a natural-language functional specification, then (2) `sp13 text2spl` generates `.spl` source from that specification, using the cookbook’s indexed recipe pairs as few-shot examples. Applying this pipeline to the PocketFlow open-source cookbook produced 42 migrated recipes (`cookbook-pocketflow/`), spanning patterns including supervisor agents, agentic RAG, MCP tool integration, self-healing workflows, text-to-SQL, Agent-to-Agent (A2A) protocol, and async batch processing. All 42 pass `sp13 validate`’s structural check (parse and I/O well-formedness), but none have yet been validated for behavioral equivalence against their Python originals on sample inputs; that validation and promotion to the main cookbook is planned as a near-term priority. The migration demonstrates a practical path to growing the SPL recipe registry organically: any sufficiently documented LLM orchestration codebase — regardless of the source framework — can be translated to `.spl` without manual rewriting.

6 Empirical Evaluation

6.1 Experiment Design

We evaluate SPL’s deterministic integration using a controlled dual-arm experiment on symbolic mathematics. The experiment covers verifier rungs R1 (SymPy) and R2 (SageMath); R3 (Lean 4) is demonstrated as a design contribution with passing unit tests but is not included in this grid (see C4 in §1.3). The benchmark consists of 20 problems spanning six difficulty tiers (T0–T5), split across two symbolic backends:

Tier	Backend	Category	Example
T0	SymPy	Single-step polynomial	$d/dx (x^4 - 2x^2 + 1)$
T1	SymPy	Multi-step polynomial chains	expand $\rightarrow$ factor $\rightarrow$ solve
T2	SymPy	Transcendental / limits / series / trig	$\sin(x)/x$ as $x \rightarrow 0$; Taylor $\sin(x)$ deg 5
T3	Sage	Integration / linear systems / eigenvalues	$\int \sqrt{4 - x^2} dx$; eigenvalues of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
T4	Sage	Laplace transforms / ODEs / summation / roots	$\mathcal{L}\{e^{-2t}\}$; $y' = y$, $y(0) = 1$; $\sum 1/n^2$
T5	Sage	Expert ODE + transform verification	$y'' - 3y' + 2y = 0$; $\mathcal{L}^{-1}\{s/(s^2 + 4)\}$ + verify

T0–T2 problems (10 total) route to the SymPy kernel; T3–T5 (10 total) route to the SageMath kernel. The backend is a per-problem parameter; the `.spl` workflow is identical across both. **Main-text scope.** To keep the primary model-vs-model comparison free of the backend-familiarity confound noted in review (a model’s SymPy-vs-Sage gap conflates API familiarity with intrinsic tier difficulty, since T0–T2 and T3–T5 are not the same problems on different backends), §6.2 reports SymPy-backend results (T0–T2) only. The

identical protocol applied to the Sage-backend problems (T3–T5) is reported in Appendix K as a reference point, not as part of the main model-capability claim.

Eleven models were evaluated: `sonnet-4-6` (Anthropic, cloud API), `gemma3` and `gemma4:e2b` (Google, local via Ollama), `qwen2.5` and `qwen3` (Alibaba, local), `deepseek-v2:16b` (DeepSeek, local), `phi3` and `phi4` (Microsoft, local), `llama3.2` (Meta, local), `l1m2.5` (Liquid AI, local), and `rnj-1` (Essential AI [25], an 8B dense Transformer based on Gemma 3, specialized for STEM and coding, local via Ollama). `qwen3` thinks by default; every `qwen3` call in this experiment carries a prompt instruction suppressing the thinking trace (“Disable thinking mode. Answer directly, with no `<think>` reasoning trace.”), since the plan stage this experiment measures is a format-translation task, not a reasoning task (see Appendix H). Each of the eleven models ran all 20 problems under two arms:

- **Solver arm** (`enable_solver=true`): the model decomposes the problem into `expr|op` steps; the symbolic kernel (SymPy or Sage) executes each step exactly; the verified chain is fed back to the model for narration. Pass = status `complete` (all steps kernel-verified end-to-end). **This pass criterion is execution-verified, not independently semantically verified:** it certifies that every decomposed step ran without error in the symbolic kernel, not that the resulting answer was checked against a known-correct value for that problem. The two are distinct — a chain can execute cleanly while still encoding an incorrect decomposition of the original problem. Independent ground-truth checking via a registered `verify(result, problem)` predicate is a separate SPL mechanism (`ASSERT`, demonstrated on the constraint-optimization and code-synthesis domains in §7.5; implementation patterns in Appendix I) and is not the pass oracle used in this experiment. **We quantify this gap directly.** A post-hoc round-trip re-verification — substituting each Pattern-1-pass result back into the original problem’s defining relation, detailed as Pattern 2 in Appendix I — was run against every solver-arm row scored pass across both runtimes (Python and Go, §5): of 3,313 Pattern-1-pass rows, **87.1% are also round-trip verified, 8.3% fail round-trip despite passing Pattern 1** (a genuine false-positive rate on the execution-only criterion), and 4.6% are unparseable logging artifacts excluded from the count. A fair, deduped, apples-to-apples comparison using the enhanced (round-trip-instrumented) recipe on both runtimes at matched scale narrows this further to 88.3% (Python) and 88.8% (Go) verified (§5). We report this here, at first mention of the pass criterion, rather than only in Appendix I, because it is the single most load-bearing correction to the pass rates reported below.
- **LLM-only arm** (`enable_solver=false`): the model solves and narrates directly with no kernel involvement. Pass = non-empty response returned (status `complete` or `unverified_success`). This arm measures *output production* — the model’s ability to generate a complete response — not mathematical correctness. Mathematical accuracy is deliberately left unverified: the comparison isolates what SPL’s deterministic mode adds (machine-verified correctness) against a baseline that represents what the LLM produces on its own.

Five sessions were run: a pilot ($r=1$, 400 cells, ten models), a repeated run ($r=3$, 1200 cells, the same ten models), a backend-split top-up ($r=3$ more, 1320 cells: 660 SymPy-problem cells + 660 Sage-problem cells, all eleven models) bringing every model to $r=6$ on both backends, an initial `qwen3` addition ($r=3$, 120 cells, thinking enabled) evaluating `qwen3` — a model not part of the original pilot/repeated sessions, added after the fact once it became clear it could complete the battery reliably — and a final `qwen3` re-run ($r=6$, 240 cells) with `qwen3`’s default thinking trace suppressed via prompt instruction, which supersedes the initial addition entirely (Appendix E, Table E.1). Session identifiers are given in Appendix E, Table E.1. Results below (§6.2) pool all $r=3$ batches restricted to the ten SymPy problems (1320 cells, eleven models, $r=6$); the matching Sage-problem pool (1320 cells, $r=6$) is reported in Appendix K. The pilot (ten models only, no `qwen3` data, $r=1$) is retained in Appendix E for comparison. The 80-recipe cookbook, experiment runner, and raw result CSVs for all sessions are available in the project repository (see Conclusion).

6.2 Results

Pass rate and latency by model and arm, SymPy backend only (T0–T2) (mean over 6 runs/cell, sorted by solver pass rate; 95% bootstrap CI on solver pass rate, 10k resamples). Sage-backend (T3–T5) results at the same $r=6$ scale are reported separately in Appendix K, per the main-text scope note above:

Model	Pass rate (%)				Latency		
	LLM-only	Solver	95% CI	Δ	LLM-only	Solver	Δ
qwen3	97	100	[100, 100]	+3	12.7s	22.0s	+74%
sonnet-4-6	100	98	[95, 100]	−2	10.2s	8.1s	−21%
gemma4:e2b	100	98	[95, 100]	−2	10.6s	11.9s	+12%
rnj-1	100	97	[92, 100]	−3	7.6s	6.6s	−12%
gemma3	100	93	[87, 98]	−7	4.1s	4.6s	+12%
phi4	100	92	[83, 98]	−8	22.0s	12.4s	−44%
qwen2.5	100	90	[82, 97]	−10	7.5s	3.8s	−49%
llama3.2	100	77	[65, 87]	−23	4.1s	2.7s	−33%
lfn2.5	77	53	[40, 65]	−23	6.8s	13.0s	+92%
deepseek-v2:16b	100	40	[28, 53]	−60	9.8s	5.2s	−47%
phi3	100	32	[20, 43]	−68	6.9s	3.4s	−51%

LLM-only “pass” = non-empty response (unverified). Solver “pass” = complete symbolic chain (machine-verified by the SymPy kernel). The two arms measure different things: output production vs. verified correctness; do not compare pass rates across columns as accuracy estimates. **Latency caveats:** (1) solver-arm latency includes early exits on `solver_error`, so lower latency for lower-scoring models partly reflects *failing faster*, not genuine speedup; a cleaner comparison would use passing runs only. (2) `sonnet-4-6` latency is measured via `claude_cli` (shell invocation) while local models use direct HTTP; cross-model latency comparisons are not instrument-normalized.

F1 — Output-fence cleaning eliminated `plan_format_error` entirely. In a preliminary run, `phi4` produced its decomposition plans wrapped in markdown fences (````plaintext ... ````), causing SyntaxErrors in the kernel and 0% solver pass. Adding `strip_fences()` — a regex-based fence stripper applied immediately after `decompose_problem()` — resolved this. Across the 400-cell pilot, the 1200-cell repeated run, the 1320-cell backend-split top-up (SymPy half reported here; Sage half in Appendix K), and the 240-cell `qwen3` re-run, zero failures were attributed to `plan_format_error` in any model. The dominant failure mode is `solver_error` (kernel-rejected expressions): *surface* format compliance (markdown fences, whitespace) is not the bottleneck — *semantic* expression correctness at the kernel boundary is. Models that fail are not failing to produce `expr|op`-shaped text; they are producing expressions the symbolic engine cannot evaluate (wrong variable names, unsupported operations, malformed syntax).

F2 — `solver_error` is the primary failure mode. Every failure in the solver arm is `solver_error` or `plan_error` — never `plan_format_error`. Models that fail are generating syntactically valid `expr|op` plans but producing expressions the kernel cannot evaluate: wrong variable names, unsupported operations, or malformed expression syntax (e.g., `^` instead of `**`). `lfn2.5` is the outlier on the SymPy set: a majority of its solver failures are `plan_error` (invalid step structure), not `solver_error`, indicating a different failure mode — plan generation rather than expression evaluation.

F3 — `qwen3` achieves the highest solver-arm pass rate as a format-translation task, thinking suppressed. `qwen3` scores 100% solver vs 97% LLM-only on the SymPy set — the only model with zero solver-arm failures at $r=6$, and higher than its own LLM-only rate. Every `qwen3` call carries a prompt instruction disabling its default thinking trace (§6.2); the plan stage `qwen3` is asked to perform — mapping a problem to `expr|op` format — does not require extended deliberation to execute reliably. `sonnet-4-6` and `gemma4:e2b` tie for second at 98%. The 95% bootstrap CIs for the top four models (`qwen3` [100, 100], `sonnet-4-6` [95, 100], `gemma4:e2b` [95, 100], `rnj-1` [92, 100]) overlap substantially, so this ranking should be read as indicative, not statistically significant at this sample size. Per-model ranking stability across the pilot

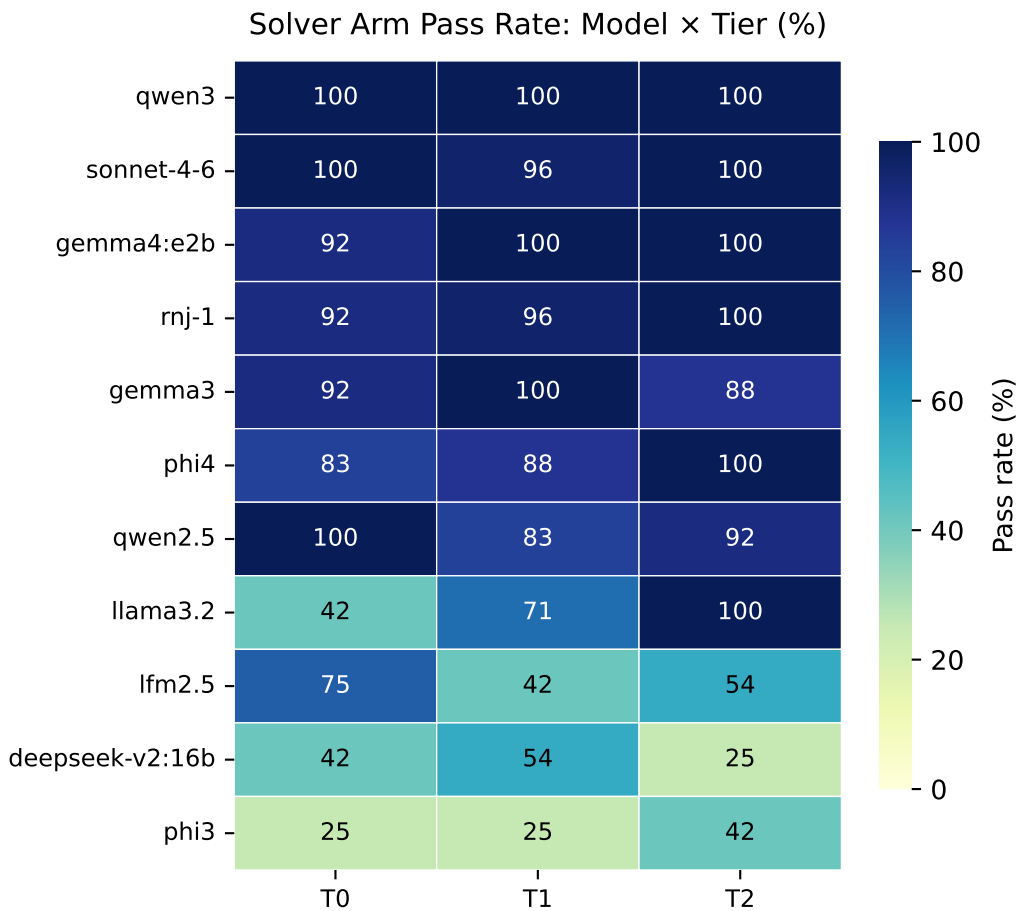


Figure 2: Solver Arm Pass Rate, SymPy backend only (T0–T2): Models (rows, sorted by aggregate) $\times$ tiers (columns). Diagonal gradient shows capability hierarchy within a single backend, avoiding the API-familiarity confound a cross-backend heatmap would introduce (see main-text scope note, §6.1, and Appendix K for the Sage-backend counterpart).

and repeated runs is a secondary, model-evaluation question rather than a claim about SPL’s architecture; full detail is in Appendix E.5 and Appendix I.

The Sage backend (T3–T5) is not compared against SymPy here, since the two use disjoint problem sets and a pass-rate gap would conflate backend-API familiarity with tier difficulty; the identical protocol on Sage problems is reported separately in Appendix K.

The SPL status codes (`plan_error`, `solver_error`, `complete`) make per-category failure analysis automatic — each cell records its failure mode, enabling per-tier, per-backend, and per-model breakdown without log parsing.

Extended discussion of the evaluation-design choices behind these numbers — the pass/fail asymmetry between arms, why scoring LLM-only accuracy is itself hard, the human-verified subsample (E1), ranking stability across repeated runs, and the thinking-mode exclusion — is provided in Appendix I to keep this section focused on results; see §8.

7 Discussion

7.1 Completing the Computation Matrix

Table 2 situates SPL against existing systems on the computation matrix from Section 1:

System	Probabilistic (design)	Probabilistic (runtime)	Deterministic (design)	Deterministic (runtime)
LangGraph	✓	✓	—	—
AutoGen	✓	✓	—	—
DSPy	✓	✓	—	—
AlphaGeometry	—	✓	✓	✓(geometry only)
SPL	✓	✓	✓	✓(Python-callable verifier)

No prior workflow language fills all four cells outside a single hard-coded domain. AlphaGeometry fills all four but is domain-locked to geometry. SPL fills all four for any domain with a Python-callable verifier, with the boundary between cells declared in source rather than implicit in framework internals. The programmer writes one `.spl` file; the runtime resolves which cells are active at invocation — a difference of kind, not degree. Correctness-vs-speed trade-offs become deployment decisions, not source-embedded choices.

7.2 The Probabilistic/Deterministic Decomposition

SPL’s two-mode architecture separates *probabilistic/intuitive* computation (LLM) from *deterministic/mechanical* computation (kernel). The probabilistic mode reads problems, chooses approaches, and narrates results; the deterministic mode executes algebra, checks identities, and certifies proofs. The empirical results (Finding F2 and the results table in §6.2) expose a boundary mismatch between the two: models with strong reasoning ability (100% LLM-only pass) fail at the probabilistic-mode format-mapping task SPL requires — translating problem statements into structured `expr|op` plans the kernel can evaluate. This surfaces a hierarchy: reasoning ability → format compliance → verifier access.

7.3 The Structured Output Bottleneck

The semantic format-mapping gap (Finding F2) extends beyond symbolic mathematics: any structured-output contract exposes a capability hierarchy where models lose verifier access if they cannot emit output the deterministic engine accepts, regardless of reasoning ability. A future feature could probe format compliance at runtime and route incapable models to the LLM-only arm; EVALUATE already provides the dispatch mechanism.

7.4 Pluggability vs. AlphaGeometry-Style Integration

AlphaGeometry [16] achieves state-of-the-art geometry theorem proving by hard-coding the neuro-symbolic loop: the neural model generates proof candidates, the symbolic engine (DD-Geometry) verifies them, and the system iterates. This tight coupling is appropriate for a single-domain system optimized for competition performance.

SPL takes the opposite design stance: the symbolic component is always an external, pluggable tool registered via `CREATE TOOL_API` or `@spl_tool`. This generality trades peak single-domain performance for domain-independence. A domain expert adds a new verifier in Python; the SPL workflow language and runtime require no modification. The same `ASSERT` semantics that check a SymPy polynomial identity can check a NetworkX graph property, a Z3 satisfiability result, or a scipy hypothesis test — any predicate the domain expert can express in Python. Empirically, open-source models like gemma4:e2b achieve 93% verified correctness on symbolic mathematics. The kernel, not the model, does the mathematical reasoning; the model only needs to reliably produce the structured `expr|op` format the deterministic engine requires. This is the design’s thesis: offloading mathematical reasoning to the kernel makes *format-mapping capability*, not model scale or proprietary access, the bottleneck. When a ~2B-parameter open-source model outperforms a frontier model on verified correctness, it demonstrates that the two-mode architecture makes high-quality verified computation accessible to everyone.

7.5 Domain Extension: Four Verification Oracles

The empirical evaluation in §6 focuses on symbolic mathematics (SymPy/SageMath/Lean). This section presents four additional cookbook workflows demonstrating that the `GENERATE` $\rightarrow$ `CALL solver` $\rightarrow$ `ASSERT` $\rightarrow$ `WHILE repair` pattern generalizes across domains wherever a machine-checkable correctness oracle exists. All four workflows are available in the project repository (see Conclusion).

74_concept_book — Education (knowledge graph + LLM). A single `build_concept_book.spl` workflow backed by a declarative domain graph (`{domain}_graph.yaml`) generates a complete, prerequisite-ordered textbook for any knowledge domain. The deterministic half (graph traversal, topological ordering, cycle detection via NetworkX) runs without any LLM call; the probabilistic half (prose generation per concept node) uses the LLM. Neither alone produces a coherent textbook: the LLM without the graph hallucinates prerequisite chains; the graph without the LLM produces dry node lists. The workflow has been validated on 15+ domain YAMLS (linear algebra, calculus, music theory, molecular biology, Chinese characters, and others) and generates output in any language via `-param language=zh`. It is deployed in production at <https://github.com/digital-duck/concept-book>.

78_constraint_opt — Operations Research (PuLP/CBC solver). An LLM reads a natural-language resource-allocation problem and generates PuLP (LP/MIP) solver code (`constraint_opt.spl`). The CBC solver executes and returns a JSON result with status "Optimal", "Infeasible", or "Error". `ASSERT is_optimal(@solution)` gates execution: the workflow cannot reach the interpretation step unless the solver has certified the global optimum with a proof of optimality. A repair loop feeds actual solver error messages back to the LLM for code correction. The LLM never performs arithmetic; all numerical values in the output come from the solver.

79_code_pytest — Software Engineering (pytest execution). An LLM generates a Python module from a natural-language specification (`code_pytest.spl`). A second, independent LLM call generates pytest test cases from the *same specification* — not from the code, ensuring the tests are an independent correctness oracle. `CALL run_pytest(code, tests)` executes both in a temporary subprocess. `ASSERT all_tests_passed(@pytest_result)` gates on the actual pytest exit code. A repair loop provides the exact pytest traceback to the LLM for targeted repair. This distinguishes “code that executes” from “code that satisfies its specification” at the language level — the same distinction `ASSERT` enforces in the symbolic-mathematics experiment.

76_lean_proof — Formal Mathematics (Lean 4 kernel). Already part of the verifier ladder (Rung R3, §4.4). Included here to make the cross-domain pattern explicit: an LLM formalizes a prose mathematical claim in Lean 4 syntax; a second LLM call writes proof tactics; the Lean kernel checks the proof; `ASSERT`

gates on the kernel verdict. A citation-path fast-pass (Mathlib lookup before proof attempt) runs first at zero token cost. This is the same `GENERATE` $\rightarrow$ `CALL solver` $\rightarrow$ `ASSERT` $\rightarrow$ `WHILE repair` pattern as the other three workflows, applied to the highest correctness guarantee available in the verifier ladder.

These four workflows, combined with the symbolic-mathematics experiment in §6, cover five verification domains (algebra, education, operations research, software, formal proof). In each case the `ASSERT` oracle is domain-appropriate and machine-checkable; the pattern requires no modifications to the SPL runtime to add a new domain — only new `CREATE TOOL_API` registrations, plus whatever domain-support code those tools need (see the Limitations paragraph on this below). Full benchmarks across all four domains are planned as a near-term priority.

7.6 Limitations

Kernel startup overhead. The IPython kernel adds ~2 seconds of cold-start latency per run. For short workflows with a single `SOLVE` step, this overhead is disproportionate. The `KernelSession` substrate (in-process `exec()`) is a lower-overhead alternative but lacks the full scientific Python environment of the Jupyter kernel.

Verifier ladder setup cost. R2 (SageMath) and R3 (Lean 4) require non-trivial installation beyond `pip install spl-llm`. SageMath requires a separate conda package or system install; Lean 4 requires the Elan toolchain and Mathlib download (~2 GB). These are not barriers for a practitioner setting up a proof-carrying pipeline, but they mean the full verifier ladder is not available in a zero-configuration deployment.

Domain-support code is not eliminated, only orchestration is. SPL’s claim is that the *workflow* — the sequencing of `GENERATE/EVALUATE/WHILE` calls against `SOLVE/ASSERT` checks — is expressed compactly in one declarative spec. It is not a claim that all supporting code disappears. `74_concept_book` is the clearest case: `build_concept_book.spl` (~11 KB) is the orchestration, but it is backed by several hundred lines of Python (`graph_lib.py`, `tools.py`, `style_profiles.py`) that a domain author writes once to give the workflow a graph structure and prose style to draw on — a reviewer opening that directory sees more Python than SPL by volume. `76_lean_proof` similarly depends on the framework’s `spl3/lean_bridge.py` kernel bridge. We consider this domain-support infrastructure — written once per domain or per verifier, analogous to a schema or ORM layer in a SQL application — distinct from the orchestration logic SPL claims to simplify, but we do not consider that line self-evidently closed: a future `GRAPH` or `STYLE` primitive that absorbed common graph-traversal and prose-styling patterns natively would be a more complete answer than asserting the distinction and leaving the Python in place. As written, the honest claim is narrower than “no glue code”: the orchestration is declarative and compact; the domain infrastructure it calls into is not, and current recipes still hand-roll it in Python.

Pass oracle. The `r=3` results in this paper use SPL status codes as the pass oracle: a solver-arm cell passes when the symbolic kernel executes all decomposed steps without error (status `complete`). The LLM sanity gate (an independent `sonnet-4-6` judge, described in Appendix E.6) was used as an additional validation check in the 400-run pilot, not as the primary oracle. The LLM-only arm’s pass criterion is deliberately weaker (non-empty response only) and is reported strictly as a reference baseline, not an accuracy claim; the reasoning behind that scope choice, and a low-effort path to closing it, are in Appendix I.

Latency is not instrument-normalized. Seven of eleven models are faster in the solver arm (led by `qwen2.5` at -55%), because the kernel short-circuits LLM chain-of-thought once decomposition is done. Part of this speedup is an artifact of *failing faster*: early-exit on `solver_error` skips the narration call entirely, lowering mean latency for lower-scoring models rather than reflecting genuine acceleration. A cleaner comparison would report latency on passing runs only (the table in §6.2 reports all-run means). `sonnet-4-6` latency is additionally measured through `claude_cli` (shell invocation) while local models use direct HTTP, so cross-model latency is not instrument-normalized. These are measurement caveats on a secondary metric, not a claim about SPL’s architecture.

7.7 Future Work

Format-compliance routing. A pre-flight probe checking whether the active model can emit the required format would gate the solver arm and eliminate format-error failures.

Cross-arm correctness verification. The LLM-only arm currently reports output production, not correctness (Appendix I). Because `enable_solver` is a single boolean parameter switching between arms on an otherwise identical `.spl` recipe (§6.1), extending the round-trip `verify(result, problem)` check already used for the solver arm to the LLM-only arm requires no new workflow, only applying the existing check unconditionally to both arms’ outputs. This makes a true apples-to-apples correctness comparison between arms straightforward future work, not a redesign.

Query optimizer. SPL’s declarative semantics enable a runtime optimizer: merging redundant `GENERATE` calls, caching `SELECT` results, model downgrading for cheap sub-tasks — all without modifying the `.spl` source.

Proof-carrying workflows. R3 (Lean 4) currently generates a proof badge but does not embed the proof artifact in the workflow’s output. A proof-carrying extension would attach the Lean 4 certificate as a typed field in the `RETURN` payload, enabling downstream consumers to independently re-verify without re-running the workflow.

Multi-domain benchmarks. Full quantitative benchmarks for `78_constraint_opt` (operations research), `79_code_pytest` (software engineering), and `74_concept_book` (education) are planned, following the same dual-arm design used in the symbolic-mathematics experiment. Each domain has a machine-checkable oracle (LP optimality certificate, pytest exit code, graph acyclicity check) that enables the same `ASSERT`-based correctness measurement.

User study. A formal study measuring development effort (time-to-first-working-workflow, error rate, LOC) across SPL and imperative equivalents (LangGraph, PDL) would provide direct evidence for usability claims. This is an open priority.

Distributed verifier routing. Momagrid [22] is an open-source decentralized inference grid (GitHub repository, not peer-reviewed) that provides an SPL adapter routing LLM calls to worker nodes. As a future direction, the same routing mechanism could route `SOLVE` dispatches to verifier-specialized nodes (a node with a GPU-accelerated SageMath install, or a node with Lean 4 and Mathlib pre-warmed), enabling a distributed verifier ladder. This capability is architectural — it has not been experimentally validated in this paper.

8 Conclusion

Many orchestration frameworks today excel at the probabilistic side - LLM invocation, multi-agent routing, chain-of-thought scaffolding. The deterministic side, however, remains fragmented: symbolic computation (algebra, geometry, formal proof) lives in separate tools with no native integration path. When practitioners need both - a workflow that decomposes with an LLM and verifies the result with a solver - they resort to glue-code that is neither declarative nor portable.

SPL closes this gap through a unified declarative approach grounded in an epistemic (not temporal) distinction: the probabilistic mode (intuitive) and the deterministic mode (mechanical) are separated by their mode of computation, not their speed (§1). A single `.spl` specification expresses both modes in the same syntax, with the same variable bindings, and this separation is what enables DODA (C2, §1.3): a workflow authored once runs unchanged across Ollama (local), OpenRouter (cloud), or Momagrid (distributed).

We validate SPL through two complementary lenses. An extensive 80-recipe cookbook illustrates expressiveness across 8 workflow categories, from basic Q&A to formal proof verification, including compilation to four external orchestration targets (LangGraph, Go, TypeScript, PocketFlow). A 1320-run controlled experiment ($11 \times 10 \times 2 \times 6 = 1320$ runs on the SymPy backend; a same-scale Sage-backend experiment is reported separately in Appendix K, to avoid conflating backend familiarity with model capability) on symbolic mathematics reveals a structural finding: three models (qwen3: 100%, sonnet-4-6: 98%, gemma4:e2b: 98%)

reach the top of the solver arm’s kernel-execution pass rate. Round-trip verification isolates execution success from solved-problem correctness (§6.2); format compliance emerges as a separable skill from mathematical reasoning ability, and models that fail format translation lose verifier access entirely. The verifier ladder (SymPy for basic algebra, SageMath for advanced mathematics) makes correctness guarantees a runtime parameter, not a specification artifact; a third rung (Lean formal proof) is implemented and available via the same **ASSERT** mechanism but is not evaluated in this paper (§4).

A broader implication emerges: declarative composition fundamentally shifts the LLM’s role from mathematical reasoner to format translator. The kernel handles derivation; the LLM handles decomposition and narration. This lowers the capability bar - a small, fast, open-source model suffices when format compliance, not reasoning depth, is the bottleneck - and enables fully offline, privacy-preserving deployment via local adapters like Ollama, where user data never leaves the device yet still benefits from machine-verified correctness. A formal grammar giving both computation modes first-class status in a single specification, and a shared variable namespace that eliminates tool-schema marshaling at the mode boundary, should hold independent of the correctness-oracle questions this evaluation surfaces, and extend to other domains such as code generation and testing, scientific computation with certification, knowledge graphs with logical constraints, JSON schema validation. Source code, experimental data, and reproducibility artifacts are publicly available at <https://github.com/digital-duck/SPL.py>.

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Appendix A: Extended Backus-Naur Form (EBNF) Grammar

The grammar below covers the SPL language. Terminal symbols are quoted strings; non-terminals are lowercase identifiers. $*$ = zero or more, $+$ = one or more, $?$ = optional, $|$ = alternation. The `{@var}` interpolation syntax (used inside string expressions for kernel dispatch) is handled at the executor level and is not shown separately.

```
(* Top-level structure *)
program      ::= (import_stmt | workflow_def | procedure_def
                  | create_fn_stmt | create_tool_stmt)*

import_stmt  ::= 'IMPORT' string_literal

workflow_def ::= 'WORKFLOW' identifier
               input_decl? output_decl? 'DO'
               stmt_list
               'END'

procedure_def ::= 'PROCEDURE' identifier
                input_decl? output_decl? 'DO'
                stmt_list
                'END'

input_decl   ::= 'INPUT' ':'? param_spec (',' param_spec)*
output_decl  ::= 'OUTPUT' ':'? param_spec (',' param_spec)*
param_spec   ::= '@'identifier type_ann? ('DEFAULT' expr)?
type_ann     ::= 'TEXT' | 'LIST' | 'SET' | 'INT' | 'FLOAT' | 'BOOL'
               | 'IMAGE' | 'AUDIO' | 'VIDEO'

(* Statements *)
stmt_list    ::= stmt*
stmt         ::= generate_stmt
               | evaluate_stmt
               | while_stmt
               | exception_stmt
               | solve_stmt
               | assert_stmt
               | call_stmt
               | call_parallel_stmt
               | return_stmt
               | logging_stmt
               | assign_stmt

(* Probabilistic primitives *)
generate_stmt ::= 'GENERATE' call_expr 'INTO' '@'identifier
                 select_clause? prompt_clause? budget_clause?

budget_clause ::= 'WITH' 'OUTPUT' 'BUDGET' integer 'TOKENS'

select_clause ::= 'SELECT' select_item (',' select_item)*
select_item   ::= system_role_call
               | '@'identifier 'AS' identifier
               | string_literal 'AS' identifier
               | rag_call
```

```

| memory_call

prompt_clause ::= 'PROMPT' string_literal

system_role_call ::= 'system_role' '(' string_literal ')'
rag_call         ::= 'rag' '(' string_literal (',' expr)? ')'
memory_call      ::= 'memory' '(' string_literal ')'

evaluate_stmt    ::= 'EVALUATE' '@' identifier
                  when_clause+ else_clause? 'END'
when_clause      ::= 'WHEN' condition 'THEN' stmt_list
else_clause      ::= 'ELSE' stmt_list
condition        ::= '=' string_literal          (* deterministic equality *)
                  | string_literal              (* semantic / LLM-judge *)
                  | 'contains' '(' string_literal ')'
                  | 'startswith' '(' string_literal ')'
                  | python_expr                  (* deterministic Python bool *)

while_stmt       ::= 'WHILE' condition 'DO' stmt_list 'END'

exception_stmt   ::= 'EXCEPTION' exception_clause+ 'END'
exception_clause ::= 'WHEN' exception_type 'THEN' stmt_list
exception_type   ::= 'HallucinationDetected' | 'ContextLengthExceeded'
                  | 'BudgetExceeded' | 'ModelRefused' | 'TimeoutError'
                  | 'WorkflowCompositionError' | identifier

(* Deterministic primitives *)
solve_stmt      ::= 'SOLVE' '@' identifier type_ann? kernel_hint? ':' python_expr
kernel_hint     ::= 'SYMPY' | 'SAGE' | 'LEAN' | 'PYTHON'

assert_stmt     ::= 'ASSERT' python_expr
                  ('OTHERWISE' stmt_list)?

(* Workflow composition *)
call_stmt       ::= 'CALL' call_expr 'INTO' '@' identifier
call_expr       ::= identifier '(' arg_list? ')'
arg_list        ::= arg (',' arg)*
arg             ::= '@' identifier | string_literal | named_arg
named_arg       ::= identifier '=' expr

call_parallel_stmt ::= 'CALL' 'PARALLEL' call_stmt+ 'END'

(* Output and side effects *)
return_stmt     ::= 'RETURN' expr
                  ('WITH' identifier '=' expr (',' identifier '=' expr)*)?
                  (* COMMIT is a deprecated alias for RETURN *)

logging_stmt    ::= 'LOGGING' expr ('LEVEL' log_level)?
log_level       ::= 'INFO' | 'DEBUG' | 'WARNING' | 'ERROR'

assign_stmt     ::= '@' identifier ':' expr

(* Tool and function registration *)
create_fn_stmt  ::= 'CREATE' 'FUNCTION' identifier

```

```

    '(' param_list? ')'
    ('RETURNS' | 'RETURN') type_ann
    'AS' '$$' string_literal '$$'

create_tool_stmt ::= 'CREATE' 'TOOL_API' identifier
                  '(' param_list? ')'
                  ('RETURNS' | 'RETURN') type_ann
                  'AS' 'PYTHON' '$$' string_literal '$$'

param_list      ::= param_spec (',' param_spec)*

(* Expressions *)
expr           ::= '@' identifier
                | string_literal
                | fstring_literal
                | number_literal
                | bool_literal
                | none_literal
                | list_literal
                | map_literal
                | binary_expr
                | unary_expr
                | call_expr

fstring_literal ::= 'f"' (char | '{' '@' identifier '}')* '"'
list_literal   ::= '[' (expr (',' expr)*)? ']'
map_literal    ::= '{' (string_literal ':' expr (',' string_literal ':' expr)*)? '}'
binary_expr    ::= expr ('+' | '-' | '*' | '/' | '==' | '!=' | '<' | '>' | '<=' | '>='
                        | 'AND' | 'OR') expr
unary_expr     ::= 'NOT' expr
python_expr    ::= string_literal (* dispatched to kernel *)

(* Lexical *)
identifier     ::= [a-zA-Z_][a-zA-Z0-9_]*
string_literal ::= '"' [^"\\]* '"' | "'" [^'\\]* "'"
                | "r'" [^']* "'" | "r'" [^"]* '"'
                | "'''" char* "'''" | "''''" char* "''''"
number_literal ::= [0-9]+ ('.' [0-9]+)?
bool_literal   ::= 'true' | 'false' | 'True' | 'False'
none_literal   ::= 'None' | 'null'
comment        ::= ('--' | '#') char* newline
                | '/*' char* '*/'

```

String literals use double quotes as preferred, single quotes are accepted, but can interfere with keyword highlighting in the VS Code syntax highlighter. The `{@var}` interpolation in string literals sent to `SOLVE/ASSERT` is resolved by the executor before kernel dispatch.

python_expr is opaque to the SPL grammar. As the production above shows, `python_expr` is defined as a `string_literal` — the SPL parser does not parse the expression text itself, only the string literal that carries it to the kernel. The deterministic mode’s expressions are therefore validated at kernel-execution time, not at SPL parse time; a syntactically valid SPL program can still fail with a kernel-side `SyntaxError` or `NameError` if the interpolated expression is malformed. This is a pragmatic choice — it lets `SOLVE/ASSERT` forward arbitrary kernel-native syntax (SymPy, SageMath, or Lean) without SPL reimplementing each kernel’s grammar — but it means the “formal grammar” claim applies to workflow structure, not to the deterministic-mode payload.

Comments are stripped before parsing. Three styles are supported: line comments with `--` or `#` (until end of line), and block comments with `/* ... */` (multi-line).

Appendix B: Adapter System

B.1 Mandatory Adapters

Four adapters are always available after `pip install spl-llm`. All support `--llm <adapter>:<model_id>` format. The legacy `--adapter` and `--model` flags remain available for backward compatibility.

Adapter:Model spec	Protocol
<code>ollama:gemma3</code>	HTTP POST to <code>localhost:11434/v1/chat/completions</code> (OpenAI-compatible)
<code>claude_cli:sonnet-4-6</code>	Shells out to <code>claude</code> CLI; captures stdout
<code>openrouter:qwen/qwen3-8b</code>	HTTPS to <code>api.openrouter.ai</code> ; requires <code>OPENROUTER_API_KEY</code>
<code>momagrid:gemma3</code>	POST <code>/tasks</code> on <code>MOMAGRID_HUB_URL</code> ; poll <code>/tasks/{id}</code> for results

`gemma` and `qwen` are used as default models throughout this paper for their continuous open-source support.

B.2 dd-llm Bridge (Cloud Providers)

When the `dd-llm` package is installed, ten additional cloud providers are available through a unified bridge adapter. The bridge normalizes provider-specific authentication and response shapes behind the same `BaseAdapter` interface:

Provider	Model prefix example
Anthropic	<code>anthropic/claude-sonnet-4-6</code>
OpenAI	<code>openai/gpt-4o</code>
Google	<code>google/gemini-2.0-flash</code>
Mistral	<code>mistral/mistral-large</code>
Cohere	<code>cohere/command-r-plus</code>
Together AI	<code>together/meta-llama/Llama-3-70b</code>
Groq	<code>groq/llama3-70b-8192</code>
Fireworks	<code>fireworks/accounts/fireworks/models/mixtral-8x22b</code>
Perplexity	<code>perplexity/sonar-large-32k-online</code>
DeepSeek	<code>deepseek/deepseek-chat</code>

B.3 Adapter Bootstrap Protocol

`spl3/adapters/_init_.py` resolves adapters in three-step priority order:

1. **dd-llm bridge** — if installed, registered first for all cloud model prefixes.
2. **Bespoke fallbacks** — `claude_cli` and `openrouter` registered as standalone adapters for environments without `dd-llm`.
3. **Always-available** — `ollama` and `momagrid` always registered last as unconditional fallbacks.

B.4 Two-Method Interface

All adapters implement:

```
class BaseAdapter:
    def generate(
        self, prompt: str, system: str, model: str, **kw
    ) -> GenerationResult: ...

    def generate_multimodal(
        self, content: list[dict], system: str, model: str, **kw
    ) -> GenerationResult: ...
```

The executor dispatches to `generate_multimodal()` when any `INPUT` variable carries an `IMAGE`, `AUDIO`, or `VIDEO` type annotation; otherwise `generate()`. `GenerationResult` carries `content: str`, `model: str`, `latency_ms: float`, and `token_counts: dict`.

B.5 Momagrid Response Protocol

The Momagrid hub returns a nested result object that must not be flattened:

```
{
  "state": "COMPLETE",
  "result": {
    "content": "...",
    "latency_ms": 342,
    "agent_name": "worker-node-07"
  }
}
```

`agent_name` is load-bearing for multi-node routing and audit logging.

Appendix C: Verifier Ladder — Setup and Configuration

C.1 Rung R1: SymPy (zero additional setup)

SymPy ships with `spl-llm`:

```
pip install spl-llm          # SymPy included
spl3 run workflow.spl --kernel --llm ollama:gemma3
```

The `--kernel` flag starts an IPython kernel with `python3` kernelspec. SymPy operations are registered as SPL tools in `cookbook/67_symbolic_math/tools.py` and are available to any workflow that imports that file or registers equivalent tools.

C.2 Rung R2: SageMath

SageMath requires a separate install due to its size (~1 GB):

```
# Option A -- conda (recommended)
conda install -c conda-forge sagemath

# Option B -- system package (Ubuntu/Debian)
sudo apt install sagemath

# Register the SageMath Jupyter kernelspec
sage -python -m sage.repl.ipython_kernel.install

# Run with SageMath kernel
spl3 run workflow.spl --kernel --kernel-name sagemath --llm ollama:gemma3
```

SPL discovers the kernel via `ensure_kernelspec("sagemath")` in `spl3/kernel.py`. The same `SOLVE/ASSERT` primitives work unchanged; only the execution environment differs.

C.3 Rung R3: Lean 4 with Lean REPL

Lean 4 and the REPL are installed via the one-shot setup script:

```
# Full install: elan + Lean 4 + leanprover-community/repl (pinned v4.30.0)
bash cookbook/tools/lean/setup_lean.sh

# Optionally with Mathlib (~2 GB download, required for library search)
bash cookbook/tools/lean/setup_lean.sh --with-mathlib
```

The script installs `elan` (Lean version manager) to `$ELAN_HOME` (default `/opt/lean`) and builds the REPL with `lake`. The SPL Lean bridge (`spl3/lean_bridge.py`) wraps the REPL in a persistent session:

```
from spl3.lean_bridge import LeanREPL

repl = LeanREPL().start()          # stdlib only
repl = LeanREPL.mathlib().start()  # with Mathlib imports
```

The bridge passes 15/15 unit tests in `tests/test_lean_bridge.py`, covering statement checking, proof verification, `exact?` citation, timeout recovery, and REPL restart.

```
# Verify installation
pytest tests/test_lean_bridge.py -v

# Run recipe 76 (the full proof pipeline)
spl3 run cookbook/76_lean_proof/lean_proof.spl --kernel --llm claude_cli
```

C.4 Environment Variables

Variable	Default	Purpose
<code>MOMAGRID_HUB_URL</code>	<code>http://localhost:9000</code>	Momagrid hub endpoint
<code>OPENROUTER_API_KEY</code>	—	OpenRouter authentication
<code>ELAN_HOME</code>	<code>/opt/lean</code>	Lean toolchain root
<code>SPL_LEAN_REPL_PATH</code>	auto-detected	Path to <code>lean_repl</code> binary
<code>SPL_KERNEL_TIMEOUT</code>	120	Kernel execution timeout (seconds)
<code>SPL_WHILE_MAX_ITER</code>	15	Default WHILE loop guard

Note on `SPL_WHILE_MAX_ITER`: Complex workflows with multiple refinement loops (e.g., formalization + proof repair in Lean, or multi-step verification chains) may require higher iteration limits. If you encounter `MaxIterationsReached` exceptions, increase this value (e.g., `export SPL_WHILE_MAX_ITER=25`) to allow nested loops to complete without premature exit. The default of 15 balances safety against infinite loops with flexibility for most use cases.

Appendix D: SPL Cookbook Recipe Catalog

The cookbook contains 80 recipes (00–79) organized across 8 categories, developed iteratively as the project evolved. Each recipe demonstrates a specific pattern or construct. Active recipes run in the batch runner (`python cookbook/run_all.py`).

Purpose: This recipe collection serves two roles. (1) **Feature verification:** each recipe validates a distinct SPL language construct or adapter capability, enabling rapid regression testing across new runtime versions and model providers. (2) **Code-RAG training data:** the description/source pairs are indexed and retrieved by the `text2spl` tool to guide LLM-based code generation, improving quality through in-domain examples. Community contributions are welcomed; recipes that add new constructs or clarify existing patterns strengthen both validation and the retrieval index.

Cross-framework migration. As a further validation of DODA, we migrated 42 recipes from the open-source PocketFlow cookbook [26] to SPL using the four-stage pipeline: (S1) `splc describe` generates a plain-English functional spec from the Python source; (S2) `text2mmd` produces a Mermaid flowchart; (S3) `mmd2spl` emits an `.spl` workflow; (S4) human review and `spl3 validate` confirm the recipe parses and its declared input/output contract is well-formed. All 42 migrated recipes pass this structural validator (`python cookbook-pocketflow/validate_all.py`); this checks parse and I/O well-formedness, not behavioral equivalence to the original Python (§4.6). A decentralized **workflow registry** is planned for future work, allowing teams to publish vetted recipes and share standard patterns.

D.1 Foundations (00–09)

ID	Name	Description
00	Recipe Maker	Auto-generate SPL recipes from natural language specs
01	Hello World	Basic PROMPT and WORKFLOW: single LLM call
02	Ollama Proxy	Route requests through Ollama; test local model access
03	Multilingual	Parallel LLM calls in multiple languages; vote on result
04	Model Showdown	A/B comparison of two models on identical task
05	Self-Refine	WHILE loop refinement until quality gate passes
06	ReAct Agent	Reasoning + acting: LLM thinks, then calls tools
07	Safe Generation	Content filtering and safety guardrails
08	RAG Query	Retrieval-augmented generation; query external knowledge base
09	Chain-of-Thought	Multi-step reasoning with intermediate checkpoints

D.2 Content Generation (10–19)

ID	Name	Description
10	Batch Test	Parallel inference on a batch of prompts; collect and compare
11	Debate Arena	Two agents debate a topic; third agent judges
12	Plan and Execute	LLM decomposes task into steps; executor validates each
13	Map-Reduce	Map computation across items; reduce results with aggregation
14	Multi-Agent	Choreograph multiple specialized agents toward shared goal
15	Code Review	LLM reviews code; iterative feedback loop until approved
16	Reflection	Generate response; LLM reflects on quality; iterate if needed
17	Tree of Thought	Explore multiple reasoning paths; prune low-confidence branches
18	Guardrails	Runtime validation of LLM output against constraints
19	Memory Conversation	Multi-turn dialogue with persistent context/memory buffer

D.3 Analysis and Extraction (20–29)

ID	Name	Description
20	Ensemble Voting	Run multiple models; majority vote on classification
21	Multi-Model Pipeline	Sequence models: model A $\rightarrow$ model B, pass-through
22	Text2SPL Demo	Natural language $\rightarrow$ SPL source code generation
23	Structured Output	Enforce JSON/XML schema on LLM response; validate format
24	Few-Shot Prompting	In-context learning: embed examples in PROMPT
25	Nested Procedures	Call workflows from within workflows; composition
26	A/B Test	Run same task on two variants; measure performance delta
27	Data Extraction	Extract structured fields from unstructured text
28	Support Triage	Classify support tickets; route to specialized handlers
29	Meeting Actions	Extract action items, attendees, dates from meeting transcripts

D.4 Education and Tutoring (30–39)

ID	Name	Description
30	Code Generator	Generate code from natural language specification
31	Sentiment Pipeline	Analyze text sentiment; classify as positive/negative/neutral
32	Socratic Tutor	Ask clarifying questions; guide learner to answer
33	Interview Simulator	Conduct mock interview; score responses in real-time
34	Progressive Summarizer	Iteratively summarize text to target length/complexity
35	Hypothesis Tester	LLM proposes hypothesis; test against data; iterate
36	Tool-Use	Register and call custom Python functions as tools
37	Headline News Aggregator	Fetch, summarize, and rank news headlines by relevance
38	Bedrock Quickstart	AWS Bedrock adapter: example integration
39	Vertex AI Quickstart	Google Vertex AI adapter: example integration

D.5 Cloud and Infrastructure (40–49)

ID	Name	Description
40	Azure OpenAI Quickstart	Azure OpenAI adapter: API key config and request routing
41	Human Steering	Pause for user input; incorporate feedback before proceeding
42	Knowledge Synthesis	Combine insights from multiple documents into unified summary
43	Prompt Self-Tuning	Automatically optimize prompt structure based on output quality
44	Adaptive Failover	Primary model fails $\rightarrow$ fallback to secondary model

ID	Name	Description
45	Vision to Action	Process image input; classify scene; trigger corresponding action
46	MCP Integration	Discover and invoke tools via Model Context Protocol; LLM selects tool and parameters; HTTP or local fallback
47	arXiv Morning Brief	Fetch latest papers in topic; summarize key findings
48	Credit Risk Assessment	Evaluate financial data; output risk score with explanation
49	Regulatory News Audit	Monitor regulatory news; flag relevant updates for compliance team

D.6 Multimodal (50–59)

ID	Name	Description
50	Code Pipeline	Multimodal: parse code images $\rightarrow$ OCR $\rightarrow$ execute
51	Image Caption	Generate descriptive captions for images
52	Audio Summary	Transcribe audio; summarize transcript
53	Video Summary	Extract keyframes; caption each; synthesize summary
54	Text to Image	Generate image from text prompt (DALL-E, Stable Diffusion)
55	Text to Speech	Convert text output to audio/voice
56	Text to Video	Generate video from text script (experimental)
57	Image Format Conversion	Convert image between formats (PNG $\leftrightarrow$ JPEG $\leftrightarrow$ WebP)
58	Image Restyle	Apply style transfer or filter to image
59	Audio Format Conversion	Convert audio between codecs (MP3 $\leftrightarrow$ WAV $\leftrightarrow$ M4A)

D.7 Parallel and Compilation (60–69)

ID	Name	Description
60	Voice Dialogue	Real-time speech $\leftrightarrow$ text; interactive conversation
61	Video to Audio	Extract audio track from video
62	Video to Image	Extract keyframes or create GIF from video
63	Parallel Code Review	Review multiple code files concurrently via CALL PARALLEL
64	Parallel News Digest	Fetch and summarize multiple articles in parallel
65	LLM Splc	Compile <code>.spl</code> to imperative code (Go/TypeScript)
66	Stock Analysis	Fetch price data; LLM generates trading signals with explanation
67	Symbolic Math Problem Generator	SymPy backend: algebraic problem solving (instances verified)
68	Problem Generator	Auto-generate educational problems from topic + difficulty
69	Notebook Generator	Convert <code>.spl</code> workflow to Jupyter notebook

D.8 Verifier Ladder and Concept-Book (70–79)

ID	Name	Description
70	Linear Algebra Core Concepts	Build concept graph for linalg topics; trace prerequisites
71	Linear Algebra Concept-Book	Interactive HTML learning path: definitions $\rightarrow$ theorems $\rightarrow$ proofs
72	Verify arXiv References	LLM cites paper; kernel verifies bibliographic accuracy
73	Intro Geometry Concept-Book	Concept-book for geometry: points $\rightarrow$ lines $\rightarrow$ shapes $\rightarrow$ proofs
74	Generic Concept-Book	Template for concept-books on any STEM topic
75	SageMath Solver	Sage kernel: instances SymPy cannot reach (Galois, eigenvalues, etc.)
76	Lean Proof Verifier	Lean 4 + mathlib: prove statements; kernel checks proof
77	Neurosymbolic Solver	Main experiment: unified workflow across SymPy $\rightarrow$ Sage $\rightarrow$ Lean rungs
78	Constraint Optimization	LLM formulates PuLP/MIP constraints from a natural-language problem; CBC solver certifies optimality; ASSERT gates on the solver verdict
79	Code + Pytest Verification	LLM generates code and, independently, pytest tests from the same spec; ASSERT gates on the pytest exit code; repair loop on failure

Appendix E: Experiment — Full Results and Methodology

E.1 Experimental Protocol

Four sessions were run on the same local workstation (Ollama for local models + `claude_cli` for sonnet-4-6):

Session	Cells	Repeats	ID
Pilot	400	r=1	<code>exp-20260615-073849</code>
Repeated	1200	r=3	<code>exp-20260615-191224</code>
Backend top-up (SymPy)	660	r=3	<code>exp-20260724-200503</code>
Backend top-up (Sage)	660	r=3	<code>exp-20260724-232533</code>
qwen3 (thinking disabled)	240	r=6	<code>exp-20260725-130342</code>

An initial `qwen3` addition (`exp-20260724-045127`, r=3, 120 cells) was run after the pilot and repeated sessions, once it became clear that a mandatory-extended-thinking model could complete the battery reliably on this hardware; its cells were later folded into the two backend top-up sessions’ r=3-more pooling to reach r=6 for all eleven models. That initial addition and `qwen3`’s share of both top-up sessions ran with `qwen3`’s default thinking trace active. All of `qwen3`’s data reported in §6.2 and Appendix K instead comes from a single later session (`exp-20260725-130342`) that re-ran the complete r=6 protocol — 20 problems $\times$ 2 arms $\times$ 6 repetitions, 240 cells — with the thinking trace suppressed via prompt instruction throughout (§6.2), superseding the earlier thinking-enabled `qwen3` cells entirely rather than pooling with them. The two backend top-up sessions otherwise still provide the r=3-more repetitions for the other ten models: each adds 660 cells (11 models $\times$ 10 problems $\times$ 2 arms $\times$ 3 repetitions, of which the `qwen3` share is excluded from pooling per above) restricted to one backend’s problem set. Pooling the repeated run, the SymPy top-up (`qwen3` share excluded), and the `qwen3` re-run’s SymPy cells gives the 1320-cell, r=6, SymPy-only dataset reported in §6.2; the matching Sage-problem pooling gives the 1320-cell Sage-only dataset in Appendix K. The two backend pools are disjoint in problem content and are not merged with each other.

Pass oracle: SPL status codes — solver arm: `complete` (chain kernel-verified); LLM-only arm: `complete` or `unverified_success` (non-empty response)

Database: `cookbook/77_neurosymbolic/experiment_results.db` (SQLite)

Raw execution log (r=3 session): The full per-run SPL output log for `exp-20260615-191224` is publicly available for first-hand inspection:

https://raw.githubusercontent.com/digital-duck/SPL.py/refs/heads/main/cookbook/77_neurosymbolic/logs-spl/recipe-77-log-20260615-191224.md

The dual-arm design uses a single `.spl` workflow (`symbolic_math.spl`) with `enable_solver` as a runtime parameter. The harness (`run_experiment.py`) iterates all (model $\times$ problem $\times$ solver_mode) cells sequentially, invoking `spl3 run` for each and persisting structured results to the DB. Status codes reported in the DB:

- `complete` — full chain verified by the symbolic kernel (solver arm) or non-empty LLM response (LLM-only arm)
- `unverified_success` — LLM-only arm produced output but with `steps=0`
- `solver_error` — kernel rejected one or more steps (expression evaluation failed)
- `plan_error` — plan parsed but semantically invalid (e.g., wrong step count)

E.2 Problem Set

The 20 problems span six tiers across two backends. T0–T2 use SymPy (10 problems); T3–T5 use SageMath (10 problems). Each problem has a known ground-truth answer available in the kernel but not used as the pass oracle (status codes provide it).

Tier	Backend	Count	Category	Examples
T0	SymPy	2	Poly single-step	$d/dx(x^4 - 2x^2 + 1)$; $\text{simplify}(x^2 - 1)/(x - 1)$
T1	SymPy	4	Poly multi-step	$\text{expand}(x + 1)^2 \rightarrow \text{factor}$; diff $3x^3 - x \rightarrow \text{factor} \rightarrow \text{solve}$
T2	SymPy	4	Transcendental/limits/series	$\sin(x)/x \rightarrow 0$; Taylor $\sin(x)$ $n = 5$
T3	Sage	4	Integration/systems/eigenvalues	$\int \sqrt{4 - x^2} dx$; $x + y = 5, x - y = 1$; $\text{eig}[[1,2],[3,4]]$
T4	Sage	4	Laplace/ODEs/sums/roots	$\mathcal{L}\{e^{-2t}\}$; $y' = y, y(0) = 1$; $\sum 1/n^2$; roots $x^4 - 1$
T5	Sage	2	2nd-order ODE + verify	$y'' - 3y' + 2y = 0$; $\mathcal{L}^{-1}\{s/(s^2 + 4)\}$ verify

E.3 Per-Tier Pass Rates (r=3 session, exp-20260615-191224)

Solver arm pass rate by model and tier (%) — mean over 3 runs/cell:

Model	T0	T1	T2	T3	T4	T5	Overall
gemma4:e2b	100	100	100	100	75	83	93
sonnet-4-6	100	92	100	100	58	50	85
rnj-1	83	100	100	50	92	50	82
gemma3	83	100	75	75	50	50	73
qwen2.5	100	100	92	67	33	33	72
phi4	67	75	100	50	50	50	67
llama3.2	67	83	100	42	42	50	65
deepseek-v2:16b	50	75	50	58	42	33	53

Model	T0	T1	T2	T3	T4	T5	Overall
lfm2.5	67	33	25	50	25	33	37
phi3	33	33	33	42	33	0	32
Tier avg	75	79	77	63	50	43	—

LLM-only arm pass rate by model and tier (%) — mean over 3 runs/cell:

Model	T0	T1	T2	T3	T4	T5	Overall
sonnet-4-6	100	100	100	100	100	100	100
rnj-1	100	100	100	100	100	100	100
qwen2.5	100	100	100	100	100	100	100
phi4	100	100	100	100	100	100	100
phi3	100	100	100	100	100	100	100
gemma3	100	100	100	100	100	100	100
llama3.2	100	100	100	100	100	100	100
deepseek-v2:16b	100	100	100	100	100	100	100
gemma4:e2b	100	100	100	92	92	100	97
lfm2.5	100	67	75	92	75	50	77

E.4 Failure Mode Breakdown (Solver Arm, r=3, 60 runs per model)

Model	complete	solver_error	plan_error	other	runs
gemma4:e2b	56	0	4	0	60
sonnet-4-6	51	8	0	1	60
rnj-1	49	11	0	0	60
gemma3	44	16	0	0	60
qwen2.5	43	17	0	0	60
phi4	40	20	0	0	60
llama3.2	39	21	0	0	60
deepseek-v2:16b	32	28	0	0	60
lfm2.5	22	11	26	1	60
phi3	19	41	0	0	60

`plan_format_error` is zero across all models and both sessions. The `other` column captures rare edge cases: `sonnet-4-6` has one `unknown` status (p012/T1, run 1) and `lfm2.5` has one `narration_error` — neither is a `solver_error` or `plan_error`. `lfm2.5` is the sole `plan_error` outlier (26/60), indicating a plan-generation failure distinct from kernel expression errors. `phi3` accumulates 41 `solver_error` failures across 60 runs, the most of any model, reflecting its difficulty naming correct Sage operations.

E.5 r=1 vs r=3 Comparison (Pilot vs Repeated Run)

The pilot session (`exp-20260615-073849`, r=1) and repeated session (`exp-20260615-191224`, r=3) ran the identical 10-model $\times$ 20-problem $\times$ 2-arm design. Solver arm pass rates (overall %):

Model	r=1 (pilot)	r=3 (repeated)	Δ
gemma4:e2b	95	93	-2
sonnet-4-6	85	85	0
rnj-1	90	82	-8

Model	r=1 (pilot)	r=3 (repeated)	Δ
gemma3	70	73	+3
qwen2.5	75	72	-3
phi4	85	67	-18
llama3.2	65	65	0
deepseek-v2:16b	45	53	+8
lfn2.5	45	37	-8
phi3	30	32	+2

Top-3 and bottom-2 rankings are stable. The largest swing is phi4 (-18 pp), revealing pilot over-estimation. sonnet-4-6 and llama3.2 are perfectly stable at 85% and 65% respectively, showing consistent solver behavior for those models. The r=3 session is the authoritative result used in §6.

E.6 Judge Prompt

The LLM sanity gate uses the following prompt for each run:

You are a mathematics judge. You will be given a problem and a proposed answer.
Determine whether the answer is mathematically correct.

Problem: {problem}

Proposed answer: {answer}

Reply with exactly one word: "pass" if the answer is correct, "fail" if it is not.
Do not explain. Do not add punctuation.

The judge model (claude-sonnet-4-6) is run independently of the experiment sessions and has no access to the SymPy ground truth.

Appendix F: Compilation Pipeline — Target Examples

The spl3 splc compiler translates a .spl workflow to idiomatic code in each target framework. All examples below compile from the same source: a minimal two-mode workflow that decomposes a math problem and verifies it with SymPy.

Source: verify_step.spl

```

WORKFLOW verify_step
  INPUT: @problem TEXT
  OUTPUT: @report TEXT
DO
  GENERATE decompose(@problem) INTO @steps
    SELECT @problem AS problem
    PROMPT "Decompose into expr|op steps, one per line: {@problem}"

  SOLVE @result SYMPY := "solve_chain({@steps})"

  ASSERT verify({@result}, {@problem})
    OTHERWISE RETURN @report WITH status = 'verification_failed'

  GENERATE narrate(@result, @problem) INTO @report
    SELECT @result AS result, @problem AS problem
    PROMPT "Explain this verified result in plain language: {@result}"

```

```
    RETURN @report WITH status = 'complete'
END
```

F.1 LangGraph Target

```
spl3 splc verify_step.spl --target langgraph

# verify_step_langgraph.py -- generated by spl3 splc
from langgraph.graph import StateGraph, END
from typing import TypedDict

class State(TypedDict):
    problem: str
    steps: str
    result: str
    report: str
    status: str

def node_decompose(state: State) -> State:
    prompt = f"Decompose into expr|op steps, one per line: {state['problem']}"
    state["steps"] = llm_call(prompt)
    return state

def node_solve(state: State) -> State:
    state["result"] = kernel_exec(f"solve_chain({state['steps']!r})")
    return state

def node_assert(state: State) -> State:
    ok = kernel_exec(f"bool(verify({state['result']!r}, {state['problem']!r}))")
    state["status"] = "ok" if ok == "True" else "verification_failed"
    return state

def node_narrate(state: State) -> State:
    prompt = f"Explain this verified result in plain language: {state['result']}"
    state["report"] = llm_call(prompt)
    return state

def route_assert(state: State) -> str:
    return "narrate" if state["status"] == "ok" else END

graph = StateGraph(State)
graph.add_node("decompose", node_decompose)
graph.add_node("solve", node_solve)
graph.add_node("assert_gate", node_assert)
graph.add_node("narrate", node_narrate)
graph.set_entry_point("decompose")
graph.add_edge("decompose", "solve")
graph.add_edge("solve", "assert_gate")
graph.add_conditional_edges("assert_gate", route_assert)
graph.add_edge("narrate", END)
app = graph.compile()
```

F.2 Go Target

```
spl3 splc verify_step.spl --target go

// verify_step.go -- generated by spl3 splc
package main

import (
    "fmt"
    "splruntime" // SPL Go runtime shim
)

type VerifyStepState struct {
    Problem string
    Steps   string
    Result  string
    Report  string
    Status  string
}

func VerifyStep(problem string) (string, error) {
    s := &VerifyStepState{Problem: problem}

    var err error
    s.Steps, err = splruntime.Generate("Decompose into expr|op steps, one per line: "+s.Problem)
    if err != nil { return "", err }

    s.Result, err = splruntime.KernelExec(fmt.Sprintf("solve_chain(%q)", s.Steps))
    if err != nil { return "", err }

    ok, err := splruntime.KernelAssert(
        fmt.Sprintf("bool(verify(%q, %q))", s.Result, s.Problem))
    if err != nil || !ok {
        return "", splruntime.CommitStatus("verification_failed")
    }

    s.Report, err = splruntime.Generate(
        "Explain this verified result in plain language: " + s.Result)
    if err != nil { return "", err }

    return s.Report, nil
}
```

F.3 TypeScript Target

```
spl3 splc verify_step.spl --target typescript

// verifyStep.ts -- generated by spl3 splc
import { generate, kernelExec, kernelAssert } from "./spl-runtime";

interface VerifyStepState {
    problem: string;
    steps?: string;
    result?: string;
    report?: string;
}
```

```

}

export async function verifyStep(problem: string): Promise<string> {
  const s: VerifyStepState = { problem };

  s.steps = await generate(
    'Decompose into expr|op steps, one per line: ${s.problem}'
  );

  s.result = await kernelExec('solve_chain(${JSON.stringify(s.steps)})');

  const ok = await kernelAssert(
    'bool(verify(${JSON.stringify(s.result)}), ${JSON.stringify(s.problem)})'
  );
  if (!ok) throw new Error("verification_failed");

  s.report = await generate(
    'Explain this verified result in plain language: ${s.result}'
  );
  return s.report;
}

```

F.4 PocketFlow Target

```

spl3 splc verify_step.spl --target pocketflow

# verify_step_pocketflow.py -- generated by spl3 splc
from pocketflow import Flow, Node
from spl_runtime import llm_call, kernel_exec # SPL PocketFlow runtime shim

class DecomposeNode(Node):
    # SPL: GENERATE decompose(@problem) INTO @steps
    def prep(self, shared):
        return shared["problem"]
    def exec(self, problem):
        return llm_call(
            f"Decompose into expr|op steps, one per line: {problem}"
        )
    def post(self, shared, prep_res, exec_res):
        shared["steps"] = exec_res
        return "solve"

class SolveNode(Node):
    # SPL: SOLVE @result SYMPY := "solve_chain({@steps})"
    def prep(self, shared):
        return shared["steps"]
    def exec(self, steps):
        return kernel_exec(f"solve_chain({steps!r})")
    def post(self, shared, prep_res, exec_res):
        shared["result"] = exec_res
        return "assert"

class AssertNode(Node):
    # SPL: ASSERT verify({@result}, {@problem}) OTHERWISE RETURN ... 'verification_failed'
    def prep(self, shared):

```

```

        return shared["result"], shared["problem"]
def exec(self, prep_res):
    result, problem = prep_res
    return kernel_exec(f"bool(verify({result!r}, {problem!r}))")
def post(self, shared, prep_res, exec_res):
    if exec_res == "True":
        return "narrate"
    shared["status"] = "verification_failed"
    return None    # end flow early

class NarrateNode(Node):
    # SPL: GENERATE narrate(@result, @problem) INTO @report
    def prep(self, shared):
        return shared["result"]
    def exec(self, result):
        return llm_call(
            f"Explain this verified result in plain language: {result}")
    def post(self, shared, prep_res, exec_res):
        shared["report"] = exec_res
        shared["status"] = "complete"
        return None

def build_flow() -> Flow:
    decompose = DecomposeNode()
    solve = SolveNode()
    assert_gate = AssertNode()
    narrate = NarrateNode()
    decompose - "solve" >> solve
    solve - "assert" >> assert_gate
    assert_gate - "narrate" >> narrate
    return Flow(start=decompose)

```

PocketFlow maps SPL constructs to its ETL pattern: `prep(shared)` extracts `@variables` from the shared store (Extract), `exec()` performs the LLM call or kernel dispatch (Transform), and `post()` writes results back and returns the next action string (Load). Deterministic nodes (SOLVE, ASSERT) call `kernel_exec()` in `exec()` rather than `llm_call()`; branching in ASSERT ... OTHERWISE is expressed as a routing return value ("narrate" or None to end early).

F.5 DODA Invariant Across Targets

The `.spl` source is unchanged across all four targets. The compiler manages:

SPL construct	LangGraph	Go	TypeScript	PocketFlow
GENERATE	StateGraph node	splruntime. Generate()	await generate()	Node.exec() → llm_call()
SOLVE	StateGraph node	splruntime. KernelExec()	await kernelExec()	Node.exec() → kernel_exec()
ASSERT ... OTHERWISE	conditional edge	if !ok branch	if (!ok) throw	post() routing: str or None
CALL PARALLEL	parallel nodes	goroutines + sync.WaitGroup	Promise.all([...])	BatchNode or parallel Flow
RETURN (COMMIT is deprecated)	END node	return / error	return / throw	None from post(); result in shared

The DODA principle holds at the compilation level: the same declarative specification produces idiomatic, correct code in each target language without any manual adaptation.

Appendix G: Runtime Implementation Details

G.1 Kernel Substrates

`spl3/kernel.py` provides two execution substrates for deterministic nodes:

`IPythonKernel` runs an out-of-process Jupyter kernel via `jupyter_client.KernelManager`. It is lazy-started on the first `SOLVE` or `ASSERT` encountered during a workflow run, then held alive for the run's duration. All state — imported modules, defined variables, intermediate results — persists across steps within a session. The kernel exposes a thread-locked `execute()` method that captures stdout and the `text/plain` repr of the last expression. Python-level errors raise `KernelExecutionError`, which the executor maps to the SPL exception hierarchy.

`KernelSession` is a lightweight in-process `exec()` substrate with a persistent namespace dict. It is used for `CREATE TOOL_API` body execution, where the overhead of an out-of-process kernel is unnecessary. Two isolation scopes are supported: `"session"` (shared namespace across the run) and `"workflow"` (fresh namespace per workflow invocation).

The kernel rung (SymPy / SageMath / Lean 4) is selected via `--kernel-name` at invocation time; setup instructions for each rung are in Appendix C.

G.2 Template Resolution and Kernel Dispatch Harness

Before a `SOLVE` expression is sent to the kernel, the executor resolves `{@var}` interpolations:

```
def _resolve_python_template(self, expr: str, state: WorkflowState) -> str:
    return re.sub(
        r'\{@(\w+)\}',
        lambda m: str(state.get_var(m.group(1))),
        expr
    )
```

The resolved expression is then wrapped in a standard harness before dispatch:

- **SOLVE:** `_spl_solve_result = {expr}; print(str(_spl_solve_result))` — printed output is captured and assigned to the target `@var`.
- **ASSERT:** `_spl_assert_result = bool({expr}); print(_spl_assert_result)` — the executor checks the output equals `"True"`; on mismatch the `OTHERWISE` body executes.

This wrapping is invisible to the workflow author: `SOLVE` and `ASSERT` behave as if Python expressions natively return SPL variables.

G.3 Adapter Bootstrap and Momagrid Protocol

The adapter bootstrap order in `spl3/adapters/__init__.py` is three-step:

1. **dd-llm bridge** — if installed, registered first for all cloud model prefixes (Anthropic, OpenAI, Google, Mistral, Cohere, Together, Groq, Fireworks, Perplexity, DeepSeek).
2. **Bespoke fallbacks** — `claude_cli` and `openrouter` registered as standalone adapters.
3. **Always-available** — `ollama` and `momagrid` registered unconditionally.

The Momagrid adapter submits tasks via `POST /tasks` and polls `GET /tasks/{id}`. The hub response carries a nested result object:

```
{"state": "COMPLETE", "result": {"content": "...", "latency_ms": 342, "agent_name": "worker-07"}}
```

`agent_name` is load-bearing for multi-node routing and audit logging; it must not be flattened into the top-level response.

G.4 Ecosystem Tooling

`spl3 text2spl` converts a natural-language description to valid `.spl` source using Code-RAG: the 70+ cookbook recipes are indexed as description/source pairs and retrieved by similarity. A validation loop parses the generated output, feeds parse errors back to the LLM, and iterates until syntactically valid.

`spl3 vibe` performs one-shot NL-to-working-code generation: natural language $\rightarrow$ `.spl` workflow $\rightarrow$ runnable Python implementation + README + test data. The `--out-dir` flag writes all artifacts to a folder; `--adapter` selects the generating model.

Both tools are available after `pip install spl-llm` with no additional configuration.

G.5 Lean 4 Four-Stage Protocol (SPL Source)

The full SPL implementation of the Lean rung referenced in §4, including the faithfulness-judgment stage (Stage 3 below), as shipped in `cookbook/77_neurosymbolic/symbolic_math.spl`. The faithfulness judge is not a built-in SPL primitive, only an ordinary `GENERATE` call:

```
-- Initialise the Lean REPL (once per session)
CALL run_python("from spl3.lean_bridge import LeanREPL;
                 _spl_lean = LeanREPL.mathlib().start();
                 print('ready')") INTO @lean_status

-- Stage 1: Formalize
GENERATE formalize_claim(@problem) INTO @lean_stmt
@check_code := f"print(_spl_lean.statement_ok(r'''{@lean_stmt}'''))"
CALL run_python(@check_code) INTO @stmt_ok

-- Stage 2: Typecheck with self-repair loop
@tries := 0
WHILE @tries < @max_tries DO
  EVALUATE @stmt_ok
  WHEN = "False" THEN
    GENERATE fix_formalization(@problem, @lean_stmt, @lean_feedback) INTO @lean_stmt
    CALL run_python(@check_code) INTO @stmt_ok
  ELSE
    @tries := @max_tries
  END
@tries := @tries + 1
END

-- Stage 3: Judge faithfulness -- a second, independent LLM call compares
-- the formalized Lean statement against the original claim. This is a
-- probabilistic judgment, not a kernel check: an unfaithful formalization
-- can typecheck (and even prove) while still misrepresenting the claim,
-- which is exactly the class of error this stage exists to catch.
@badge := "statement_checked"
GENERATE judge_faithfulness(@problem, @lean_stmt) INTO @faithful
EVALUATE @faithful
WHEN contains("UNFAITHFUL") THEN
  @badge := "unfaithful"
```

```

    RETURN @lean_stmt WITH status = @badge
  ELSE
    DO NOTHING
  END
END

-- Stage 4: Prove with self-repair loop
GENERATE write_proof(@lean_stmt) INTO @lean_proof
@prove_code := f"print(_spl_lean.proof_ok(r'''{@lean_stmt}''', r'''{@lean_proof}'''))"
CALL run_python(@prove_code) INTO @proof_ok

@proof_tries := 0
WHILE @proof_tries < @max_tries DO
  EVALUATE @proof_ok
  WHEN = "False" THEN
    GENERATE fix_proof(@lean_stmt, @lean_proof, @lean_feedback) INTO @lean_proof
    CALL run_python(@prove_code) INTO @proof_ok
  ELSE
    @proof_tries := @max_tries
  END
  @proof_tries := @proof_tries + 1
END

RETURN @lean_proof WITH status = 'machine_proved'

```

The LLM’s role in the Typecheck and Prove stages is proposal, not certification — correctness of the *formalized statement* and its proof is decided by the Lean kernel, not by the model. A formalization that fails to compile against Mathlib is caught at Typecheck; a claim the kernel cannot derive a proof for is caught at Prove; in either case `machine_proved` is withheld. But typechecking and provability are properties of the Lean statement in isolation — they cannot detect a formalization that is well-formed *and* provable while still misrepresenting the original natural-language claim (e.g., an inequality flipped, a quantifier’s scope narrowed, a hypothesis silently dropped). That class of error is exactly what the faithfulness-judgment stage (Stage 3) exists to catch, and it is caught by a second LLM call, not by the kernel: `judge_faithfulness` is the one probabilistic gate in an otherwise kernel-certified chain. We report this plainly rather than let the badge name imply otherwise: `machine_proved` certifies that the proof is valid *for the Lean statement the faithfulness judge accepted*, not that the judge was correct to accept it.

Using a second, independent LLM call for this task is a defensible design choice — LLMs are comparatively well-suited to exhaustive, fatigue-free comparison of two texts for a mismatch, a proofreading-style task where human review is known to degrade from skimming and expectation bias — but this is a plausibility argument for the design, not a substitute for measuring it: this paper reports no independent accuracy estimate for the faithfulness judge itself, and one is needed before the claim can be strengthened further. This is what distinguishes an LLM-assisted proof pipeline from an LLM-only one: the deterministic mode’s guarantee attaches to the kernel’s verdict on a statement whose faithfulness is judged, not proven.

The cost/confidence trade-off across rungs is explicit: a SymPy algebraic check completes in milliseconds with zero external dependencies; a SageMath verification takes seconds and requires a separate conda environment; a Lean 4 formal proof may take tens of seconds to minutes (including Mathlib compilation on first use) and requires the Elan toolchain plus ~2 GB of Mathlib. The DODA principle lets the workflow author choose the appropriate operating point at invocation time — routine computation uses R1, publication-quality claims escalate to R3 — without modifying the `.spl` source.

G.6 Privacy and Edge Deployment

SPL’s architecture makes a fully local, zero-egress deployment a one-flag configuration choice rather than a workaround. Any local-execution stack (including plain Python calling local models) shares this property, but

SPL preserves it without requiring the workflow author to write different orchestration code for the local and cloud cases. Ten of the eleven models in the 1,320-run experiment are open-source and run entirely locally via Ollama or llama.cpp — no data leaves the device, no API key is required, and the SymPy/SageMath kernels are equally offline. `sonnet-4-6` is the sole cloud model and is included solely as a frontier-model reference point; it is not required for any of the 80 cookbook recipes or for the solver arm itself. A practitioner who excludes it retains a fully local, zero-egress stack.

This architecture is consistent with the data-sovereignty requirements of regulations such as FERPA and GDPR: because the `.spl` specification is backend-agnostic, switching from cloud to local execution is a one-flag change (`-adapter ollama`) with no workflow modifications. No formal privacy audit or compliance review has been conducted; we make no legal guarantee of regulatory compliance. What we do claim is a *design stance*: SPL defaults to local, open-source execution and treats cloud API calls as optional, opt-in overrides. The 1,320-run experiment demonstrates this stance is practical at scale — 10/11 models, zero API cost, full machine-verified correctness.

Appendix H: Model Selection for Two-Mode Workflows

Two task types arise in a declarative two-mode workflow; they suit different model behaviors.

Property	Probabilistic	Deterministic
Goal	Pattern recognition, format mapping	Exact, reproducible derivation
Output	Approximate, context-shaped	Verifiable, same answer every run
Extended thinking	Liability — wastes tokens on work the kernel will do	N/A — correctness guaranteed by construction
Key metric	Structured output compliance	Machine-checkable correctness
Speed	Variable	Can be faster than probabilistic for structured problems
Role in SPL	LLM (GENERATE, EVALUATE)	Kernel (SOLVE, ASSERT)

Plan-and-Explain Pattern for Probabilistic Tasks

The two-mode architecture enables a reusable pattern for workflows that combine LLM reasoning with deterministic verification. The **plan-and-explain** pattern decomposes the LLM’s work into three stages:

1. **Plan** (probabilistic): The LLM breaks down the problem into a structured format the deterministic engine understands — e.g., decomposing a math problem into `expression|operation` steps, or translating a natural-language claim into a formal Lean statement.
2. **Execute** (deterministic): The kernel (SymPy, SageMath, Lean, or any Python-callable verifier) processes the plan step-by-step, producing exact, reproducible results. Errors are caught and reported explicitly via exception handling or status codes.
3. **Explain** (probabilistic): The LLM narrates the kernel’s output in natural language, providing context, interpretation, or summary. Since the result is already verified, the LLM’s role is presentation, not reasoning — it can use smaller, faster models. This stage is critical for accessibility: mathematicians and domain experts often express their work in cryptic or arcane notation (specialized symbols, terse formalism, rare terminology) that few understand. The explain step translates verified results into accessible prose, making specialized knowledge available to a broader audience — turning rare textbooks and dense proofs into comprehensible narratives without sacrificing rigor or correctness.

This pattern appears throughout the cookbook: recipe #67 (symbolic math) plans a chain of algebraic steps, verifies with SymPy, then explains the result; recipe #76 (Lean proof) formalizes a claim, checks it with

the kernel, and interprets the proof badge. The pattern generalizes: any domain with a Python-callable verifier (unit testing, schema validation, graph properties, constraint satisfaction) can use plan-and-explain to combine LLM fluency with deterministic correctness.

In the SPL neurosymbolic experiment the LLM’s job is **probabilistic**: recognize the problem type and map it to **expr|op** format (plan stage). The kernel’s job is **deterministic**: execute each step exactly (execute stage). The LLM then explains the verified result. **qwen3**, a model that thinks by default, has that trace suppressed via a prompt instruction in this experiment (§6.2) and completes the plan stage reliably, topping the solver-arm ranking — consistent with plan-and-explain reframing the plan stage as a format-translation task, not a reasoning task that would need the trace.

The broader implication: declarative composition lowers the LLM capability bar for math workflows. Without SPL, solving calculus requires a model that reasons deterministically. With SPL and the plan-and-explain pattern, it requires only a probabilistic translator — the kernel handles the derivation, and a smaller model handles the explanation.

Appendix I: The `verify()` Predicate — Implementation Notes

The `verify({@result}, {@problem})` construct shown in the `verify_step.spl` listing (Appendix F, §8) and in the worked example of §3.5 is not a built-in SPL primitive. It is a **user-registered Python callable** that the kernel invokes through the **ASSERT** mechanism. This appendix clarifies how it is implemented.

Registration

Any Python function that accepts the result and problem as arguments and returns a Boolean can serve as the pass oracle. In the SPL runtime, tool functions are registered via the **CREATE TOOL_API** declaration or by decorating them with `@spl_tool` in a companion `tools.py` module. The kernel evaluates the function and gates workflow continuation on the Boolean return value.

Three Implementation Patterns

Pattern 1 — Kernel execution status (used for the reported solver-arm pass rate in §6). In `cookbook/77_neurosymbolic/`, the pass oracle used to compute every solver-arm number reported in §6 is implicit: the **ASSERT** expression checks the *exit status* of `CALL solve_chain()` rather than an explicit equality predicate. The tool `solve_step_with_sympy()` (and its SageMath counterpart) returns `'complete'` on successful evaluation and raises `solver_error` on failure. The effective gate is:

```
ASSERT {@solve_status} == 'complete'
    OTHERWISE RETURN @result WITH status = 'solver_error'
```

In this pattern, verification and execution are the same act: the kernel either succeeds or it does not. No separate `verify()` function is required. **This is Pattern 1 throughout, uniformly, for every SymPy/Sage solver-arm run scored pass in §6, on both runtimes — there is no per-tier, per-backend, or per-runtime variation in the pass criterion.** The Lean arm (R3) is a separate arm governed by the badge protocol of §4.4 and Appendix G, not this `verify()` taxonomy.

Pattern 2 — Round-trip / back-substitution (ground-truth comparison, applied post-hoc to the T0–T5 problem set). The direct approach to ground-truth comparison is to re-derive a canonical expected answer and diff it against the result:

```
def verify(result: str, problem: str) -> bool:
    expected = compute_expected(problem)
    return simplify(parse_expr(result)
        - parse_expr(expected)) == 0
```

This form compares *answer forms*, and several of the T0–T5 problems have more than one valid form (an indefinite integral’s free constant, an ODE general solution’s arbitrarily-labeled constants, an eigenvalue list’s

ordering), which forces case-by-case equality logic per problem. The implementation actually used instead follows the classroom “check your work” method — **round-trip / back-substitution**: substitute the result back into the *original problem’s defining relation* and confirm it holds, rather than comparing its form to a precomputed answer. Concretely: a **solve** result is plugged back into the original equation and checked to evaluate to zero; an **integrate** result is differentiated and checked to reproduce the original integrand (the free constant vanishes automatically under differentiation); an ODE’s returned solution is substituted back into the ODE and checked to satisfy it identically, with the general-solution problem additionally required to retain both of its independent constants (a particular solution such as bare e^x satisfies the ODE but is not the general solution asked for). All of this is performed by SymPy itself — deterministic, with no LLM in the check.

Run post-hoc (`cookbook/77_neurosymbolic/verify_roundtrip.py`) against every SymPy/Sage solver-arm row already logged in `experiment_results.db` for these problems across both runtimes (no LLM re-run; excluding rows tagged `has_thinking='Y'`, i.e. qwen3’s superseded thinking-enabled cells, §6.1), the result is: of 3,313 Pattern-1-pass rows (4,510 solver-arm rows checked in total), **87.1% are also round-trip verified, 8.3% fail round-trip despite Pattern 1 having scored them pass, and 4.6% are unparseable** (a logging/formatting artifact excluded from the count rather than asserted as a failure). Concrete false positives recovered this way include a Laplace-transform run that transformed its own already-correct answer a second time, a first-derivative run that kept differentiating past the requested order, and a general-solution run that solved the ODE but was missing one of its two required constants. Roughly one in twelve Pattern-1 passes on this problem set does not, in fact, satisfy the problem it was asked to solve. A fair, deduped comparison at matched scale using the round-trip-instrumented recipe on both runtimes (§5) narrows the verified rate to 88.3% (Python) and 88.8% (Go), confirming the gap is a property of the pass criterion, not a runtime-specific artifact. Full per-problem and per-model detail is in `roundtrip_verification_report.md` alongside the script.

Pattern 3 — Domain predicate (general case). For non-mathematical domains, `verify()` is any domain-appropriate Boolean check — JSON schema validation, unit-test pass/fail, graph connectivity, or compiler acceptance:

```
ASSERT json.loads(@api_response) and schema_valid(@api_response)
    OTHERWISE RETURN @result WITH status = 'schema_violation'
```

Multi-Approach to Verification

The three patterns above are instances of a broader point worth stating explicitly: SPL does not commit `verify()` to one fixed verification technique, because no single technique is appropriate across domains. The choice of *how* to verify is itself part of the workflow author’s design space, and maps onto strategies long used outside of software — the same repertoire a student is taught for checking a worked solution:

- **Back-substitution / round trip** — substitute the result into the problem’s defining relation (Pattern 2 above; §4.4’s discussion of what a kernel check can and cannot catch is the same idea applied to Lean).
- **Inverse-operation check** — apply the literal inverse of the last operation performed (differentiate an integral, re-transform an inverse Laplace transform) and compare to the original input; closely related to back-substitution but not identical to it in general.
- **Special-case / limiting-case sanity check** — evaluate the result at a known-easy point, a symmetry, or a dimensional/order-of-magnitude estimate, without requiring a full inverse to exist.
- **Cross-method check** — solve the same problem by a second, genuinely different technique and require agreement; catches systematic errors that repeating the same method would not.
- **Independent second-reviewer judgment** — a second, independent evaluator (human peer review, or an LLM-as-judge call blind to the first pass’s reasoning) checks the result against the original claim; this is the `79_code_pytest` recipe’s design (an independent LLM call generates tests from the specification, not from the code) and the Lean rung’s faithfulness-judgment stage (§4.4) alike.

It is a probabilistic check when the reviewer is an LLM, and its accuracy is only as good as that reviewer’s, measured or not.

- **Mechanical / formal proof-checking** — a deterministic kernel (Lean’s typechecker and tactic checker) verifies logical validity exhaustively; this is the strongest guarantee available, but — as §4.4 discusses — it certifies the proof of the statement it is given, not that the statement faithfully represents the original claim.

These strategies are not mutually exclusive and not uniformly available: an indefinite integral admits a clean inverse-operation check; a Laplace transform admits a clean round trip; a natural-language claim’s faithfulness to its own formalization admits none of the deterministic options and is left to independent judgment. The workflow author selects the strongest verification strategy the problem actually admits, and states which one was used — this appendix’s obligation to name Pattern 1/2/3 (and, per §4.4, to name the Lean badge’s faithfulness gate as a judgment rather than a kernel check) is the general form of that same discipline: a correctness claim is only as strong as the verification method backing it, and the method should be named, not left implicit in the word “verified.”

Source Code Reference

The kernel tools used in the neurosymbolic experiment are implemented in `sympolic_tools.spl` (`solve_step_with_sympy()` at line 42, `solve_step_with_sage()` at line 192):

https:
[//github.com/digital-duck/SPL.py/blob/main/cookbook/77_neurosymbolic/sympolic_tools.spl](https://github.com/digital-duck/SPL.py/blob/main/cookbook/77_neurosymbolic/sympolic_tools.spl)

Neither function contains a dedicated `def verify(...)` body — the verifier IS the solver. Symbolic computation success is the proof of correctness; the `ASSERT` gate catches any deviation via the status code.

Extended Discussion of Evaluation Design

This subsection expands on the evaluation-design choices behind the pass rates reported in §6.2.

Pass/fail criterion and baseline asymmetry. The solver arm pass criterion is machine-verified: status `complete` means the symbolic kernel executed every decomposed step without error, producing a chain-verified result. The LLM-only arm pass criterion is deliberately weaker: status `complete` or `unverified_success` means the model produced a non-empty response. This asymmetry is by design — the experiment does not claim “the solver arm is more accurate than the LLM-only arm.” Instead, it measures the *cost of verified correctness*: given that the LLM can produce output (near-100% in the LLM-only arm), how much does adding machine-verification via the solver reduce the pass rate? The answer — from 3% (gemma4:e2b) to 68% (phi3) — reveals the format-compliance hierarchy.

Why scoring LLM-only accuracy is itself hard. A natural follow-up question is: how mathematically correct are the LLM-only outputs? We attempted post-hoc verification by extracting answers from LLM narrations and comparing against the solver arm’s verified results using SymPy `simplify`. The attempt yielded unreliable scores (3–28% match rate) — not because the models are necessarily wrong, but because LLM outputs are *representationally polymorphic*: the same correct derivative may appear as `4x³ - 4x` (Unicode), `4*x**3 - 4*x` (Python), `$4x^3-4x$` (LaTeX), or “four x cubed minus four x” (prose). No heuristic extraction reliably recovers a canonical symbolic form from free-text narration, and even human reviewers struggle to verify correctness across these representations without a computational tool.

This difficulty is not a limitation of the experiment — it is the central observation the paper makes. The solver arm eliminates representational ambiguity entirely: the kernel operates on canonical symbolic expressions and returns machine-checkable results. The fact that *scoring* LLM-only mathematical accuracy requires either a human expert or a symbolic verifier is itself evidence for the two-mode architecture SPL proposes.

To be explicit about scope: this paper makes no claim, positive or negative, about the LLM-only arm’s mathematical correctness. Its numbers are reported strictly as a reference baseline against which the solver

arm’s verification cost is measured (§6.2), not as an accuracy result in their own right. Closing this gap requires no new infrastructure, only more experiment time: the same round-trip `verify(result, problem)` check already used to re-verify solver-arm passes (Pattern 2, above) applies unchanged to the LLM-only arm’s answer, since both arms return a symbolic expression that can be substituted back into the problem’s defining relation. This is made simple by an existing design choice rather than requiring a new one: `enable_solver` is a single boolean workflow parameter, so both arms already run the identical `.spl` recipe end-to-end (§6.1) — extending verification to the LLM-only arm is a matter of applying the existing check unconditionally, not building a second evaluation pipeline. This is planned for the next experiment round.

Human-verified accuracy subsample (E1). To provide a direct accuracy estimate alongside the non-empty-output proxy, we manually reviewed 24 LLM-only arm responses (4 models $\times$ 6 tiers, run 1 only; entries drawn from the `r=3` session, Appendix E Table E.1). Of 24 responses, 19 were fully correct, 1 was partially correct (11ama3.2 T0: correct differentiation method, arithmetic slip), and 4 were wrong (gemma4:e2b T3: no output produced; 11ama3.2 T2–T4: incorrect procedure or no usable answer). Per-tier accuracy: T0 3/4 (+1 partial), T1 4/4, T2 3/4, T3 2/4, T4 3/4, T5 4/4. These results confirm the expected pattern: high accuracy on simpler tiers (T0–T2), degrading on T3–T5 where the solver arm achieves 100% machine-verified correctness on equivalent problems for models that pass the format-compliance gate. Critically, all four failing 11ama3.2 cells returned `complete` in the automated LLM-only pass-rate table, demonstrating directly that non-empty output is not a correctness proxy. The subsample log is publicly available alongside the main experiment log at the repository URL in §6.1.

Repeated runs and ranking stability. The `r=1` pilot (400 cells, ten models, both backends) and the original `r=3` repeated session (1200 cells, the same ten models, both backends — prior to the later `qwen3` addition and backend top-ups described in Table E.1) tell a consistent story at the top and bottom of the ranking but diverge in the middle. The top-3 ranking (gemma4:e2b $\rightarrow$ sonnet-4-6 $\rightarrow$ rnj-1) and bottom-2 (lfn2.5, phi3) are stable across both sessions. The middle tier shows pilot variance of up to 18 percentage points (phi4: 85% $\rightarrow$ 67%), indicating that `r=3` is necessary for reliable ranking. `qwen3`, added afterward and now evaluated at `r=6` via a dedicated thinking-disabled re-run (§6.2), has no pilot-session data point for a stability comparison; it tops the SymPy-backend `r=6` ranking reported in §6.2 at 100%. Full per-run comparison is in Appendix E.5.

Thinking-mode compatibility. `qwen3` thinks by default; every call in this experiment carries a prompt instruction suppressing that trace (§6.2), since the plan stage is a format-translation task rather than one this experiment asks the model to reason through. With thinking suppressed, `qwen3` completed the full battery reliably and scored highest of all eleven models (F3, §6.2). This experiment does not test whether `qwen3`’s native thinking mode is compatible with the `expr|op` format contract — only that the format-translation task does not require it.

This distinction reflects an architectural point, not merely a token-budget constraint. SPL intentionally offloads mathematical reasoning to the deterministic kernel, reducing the LLM’s role to that of a *format translator* — mapping natural-language problems into strict `expr|op` syntax. The empirical results are consistent with this design: gemma4:e2b, a $\sim$ 2B-parameter open-source model, achieves 93% verified correctness — not because it reasons mathematically (the kernel does that), but because it reliably produces the structured format the kernel requires. For SPL’s two-mode architecture, a fast, format-compliant model is a viable tool regardless of thinking mode; the kernel, not the LLM’s reasoning trace, performs the mathematics.

SPL is not inherently incompatible with chain-of-thought reasoning — a thinking model’s hidden reasoning trace could be captured as a distinct `GENERATE` step whose output is discarded before the structured `expr|op` extraction step. Models with *optional* thinking modes (e.g., `qwen3` with `--no-think`) would work with SPL as-is by disabling the thinking trace at invocation time.

Appendix J: EVALUATE/WHILE versus DSPy Assertions and LMQL Constraints — A Worked Contrast

Section 2 notes that DSPy and LMQL are the closest existing systems in the declarative-LLM space but differ fundamentally from SPL’s EVALUATE/WHILE constructs. This appendix shows the same self-refine task — iteratively improve a draft until a quality gate is satisfied — expressed in each system.

DSPy (Stanford, 2023)

```
class SelfRefine(dspy.Module):
    def __init__(self):
        self.draft = dspy.ChainOfThought("task -> draft")
        self.critic = dspy.ChainOfThought("task, draft -> feedback")
        self.refine = dspy.ChainOfThought("task, draft, feedback -> improved")

    def forward(self, task):
        draft = self.draft(task=task).draft
        for _ in range(3):
            feedback = self.critic(task=task,
                                   draft=draft).feedback
            dspy.Assert("approved" in feedback.lower(),
                        "critic must approve")
            draft = self.refine(task=task, draft=draft,
                                feedback=feedback).improved
        return draft
```

`dspy.Assert` is a *training-time optimization directive*. During `dspy.compile()`, the teleprompter uses assertion outcomes to select few-shot examples. At inference time, a failed assertion triggers a configurable backtracking policy (retry or exception propagation), not user-defined ELSE logic. The iteration is Python, not DSPy: the declarative boundary ends at the module interface.

LMQL (ETH Zurich, 2022)

```
argmax
  "Draft: [DRAFT]" where len(DRAFT) > 50
  "Feedback: [FEEDBACK]" where
    "approved" in FEEDBACK.lower() or len(FEEDBACK) > 20
from "gpt-4"
```

LMQL operates at the *token generation level*. The `where` clause enforces structural constraints (length, regex, whitelist) during beam search over the output token sequence. It cannot semantically evaluate “is this draft high quality?” — that requires an LLM judgment call, which is outside the token-constraint model. Iteration requires wrapping in a Python loop outside the LMQL query.

SPL

```
WORKFLOW self_refine
  INPUT: @task TEXT
  OUTPUT: @draft TEXT
DO
  GENERATE draft_prompt(@task) INTO @draft
  @iter := 0
  WHILE @iter < 3 DO
    GENERATE critique_prompt(@draft, @task) INTO @feedback
```

```

EVALUATE @feedback
  WHEN contains('[APPROVED]') THEN
    RETURN @draft WITH status = 'complete'
  ELSE
    GENERATE refine_prompt(@draft, @feedback) INTO @draft
    @iter := @iter + 1
  END
END
RETURN @draft WITH status = 'complete'
END

```

EVALUATE dispatches the condition to an LLM judge at runtime; the WHEN clause is a semantic pattern evaluated by the model, not a token-level constraint or compile-time hint. WHILE is first-class iteration with a programmer-visible exit condition. The two compose naturally: each loop body can branch to an early RETURN, modify variables, or continue, with the stopping criterion declared in the specification rather than embedded in Python control flow.

Structural Differences

Feature	DSPy / LMQL	SPL
Quality gate	Training hint (DSPy) / token constraint (LMQL)	Runtime LLM judge (EVALUATE)
Iteration	Python <code>for/while</code> outside the module	First-class WHILE construct
Else-branch on failure	Backtrack policy or exception	Explicit ELSE block
Specification boundary	Module interface or single query	Full workflow including control flow

The practical consequence: a DSPy self-refine module mixes Python control flow (the `for` loop) with DSPy calls — the iterative logic is not in the declarative specification. An LMQL query cannot express iteration within the query. An SPL workflow expresses the complete iterative refinement logic as a single declarative specification that compiles unchanged to any target framework (DODA).

Appendix K: Sage-Backend Results (T3–T5)

§6.2 scopes the main-text model comparison to the SymPy backend (T0–T2) only, because T0–T2 and T3–T5 are disjoint problem sets on different backends: a SymPy-vs-Sage pass-rate gap conflates backend-API familiarity with intrinsic tier difficulty, and no amount of caveating that confound in prose makes it a fair head-to-head comparison (§6.2). This appendix reports the identical protocol — eleven models, ten Sage-tier problems, both arms, $r=6$ — on the Sage problems (T3–T5) as a self-contained reference point. **These numbers are not intended to be read side-by-side against the SymPy table in §6.2** as a backend-difficulty claim; disentangling backend familiarity from tier difficulty would require running the same problems through both backends, which this dataset does not do.

Pass rate and latency by model and arm, Sage backend only (T3–T5) (mean over 6 runs/cell, sorted by solver pass rate; 95% bootstrap CI on solver pass rate, 10k resamples):

Model	Pass rate (%)				Latency		
	LLM-only	Solver	95% CI	Δ	LLM-only	Solver	Δ
qwen3	97	98	[95, 100]	+2	14.4s	22.6s	+57%
gemma4:e2b	95	90	[82, 97]	-5	11.4s	13.4s	+18%
sonnet-4-6	100	87	[78, 95]	-13	13.0s	9.3s	-29%
rnj-1	100	78	[68, 88]	-22	7.5s	6.4s	-14%
phi4	100	70	[58, 82]	-30	23.4s	10.1s	-57%
gemma3	100	70	[58, 82]	-30	4.6s	4.8s	+4%
qwen2.5	100	68	[57, 80]	-32	9.2s	4.2s	-54%
llama3.2	100	55	[42, 67]	-45	5.3s	3.0s	-44%
lfn2.5	80	48	[37, 62]	-32	6.5s	14.0s	+114%
deepseek-v2:16b	100	38	[27, 50]	-62	11.1s	4.8s	-56%
phi3	100	30	[18, 42]	-70	7.2s	4.6s	-37%

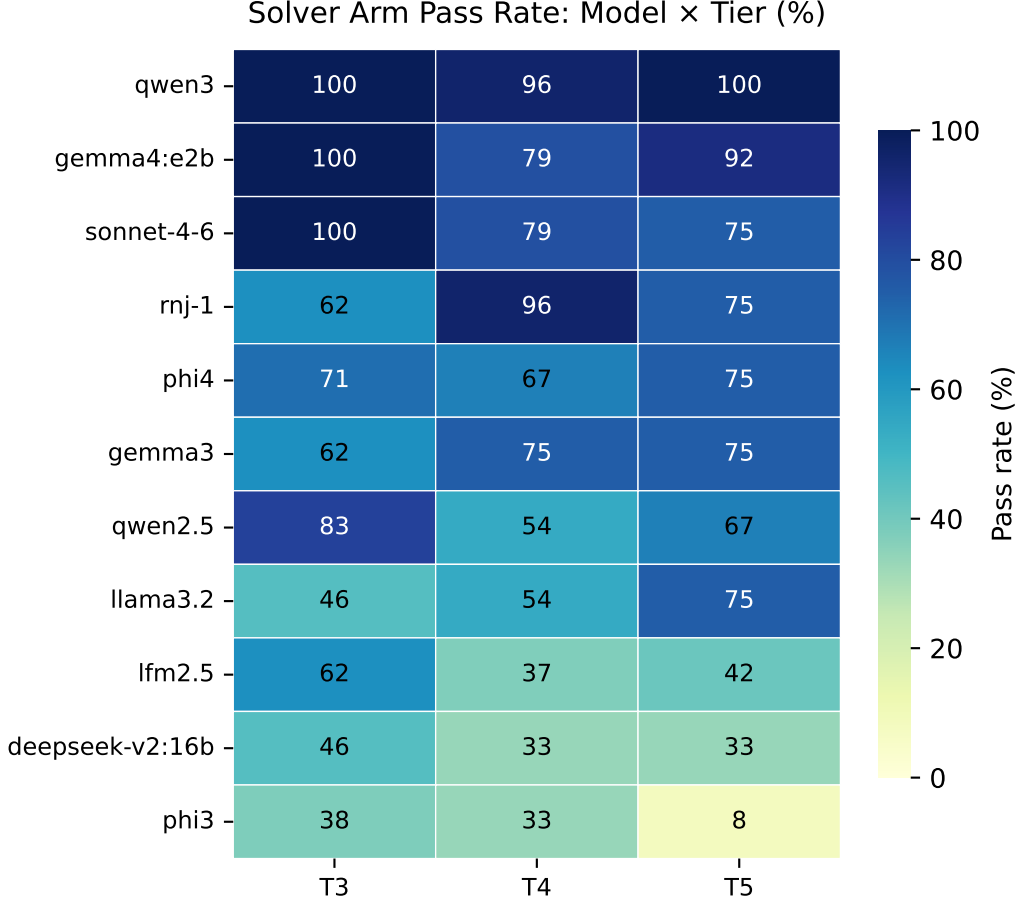


Figure 3: Solver Arm Pass Rate, Sage backend only (T3–T5): Models (rows, sorted by aggregate) $\times$ tiers (columns). Companion to Figure 2; not directly comparable to it (see scope note above).

qwen3 again leads the solver arm, and again shows one of the smallest verified/unverified gaps (98% vs 97%) — consistent with the SymPy-backend finding in F3, §6.2. The model ranking is broadly stable between the two backends (qwen3, gemma4:e2b/sonnet-4-6, and rnj-1 occupy the top four on both), while absolute pass rates are lower across the board on Sage — consistent with Sage requiring correctly-named, less-common operations (laplace, inverse_laplace, solve_system) that T3–T5’s problem content additionally

demands (Laplace transforms, ODEs, infinite sums), on top of whatever backend-familiarity gap exists. As stated above, this appendix does not attempt to separate those two effects; a matched-backend design (the same problems run through both kernels) is left as future work.